

A Novel Architecture for Collaborative Augmented Reality Experiences for Education

By Stefano Masneri

Supervised by Dr. Ana Arruarte Lasa and Dr. Mikel Zorrilla Berasategi

Donostia – San Sebastián

Friday, 8 March 2024

01

INTRODUCTION

- Motivation
- Objectives
- Methodology
- Contributions

Table of Contents

01

INTRODUCTION

- Motivation
- Objectives
- Methodology
- Contributions

02

STATE OF THE ART

- Overview
- Methodology
- Results
- Discussion

Table of Contents

01

INTRODUCTION

- Motivation
- Objectives
- Methodology
- Contributions

02

STATE OF THE ART

- Overview
- Methodology
- Results
- Discussion

03

RESEARCH RESULTS

- Interoperable architecture for collaborative AR in education
- Validation of the proposed architecture

Table of Contents

01

INTRODUCTION

- Motivation
- Objectives
- Methodology
- Contributions

02

STATE OF THE ART

- Overview
- Methodology
- Results
- Discussion

03

RESEARCH RESULTS

- Interoperable architecture for collaborative AR in education
- Validation of the proposed architecture

04

CONCLUSIONS

- Conclusions
- Publications
- Future work

1. INTRODUCTION

Scope of the research

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

AR can improve the retention
and engagement of students

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

But it has not yet seen
widespread adoption in
education

5

MOTIVATION

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

CONCLUSIONS

Conclusions

Publications

Future work

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

PROS

Wide hardware support

Software libraries

AR standards available

Different AR types

PROS

Wide hardware support

Software libraries

AR standards available

Different AR types

CONS

Single user experiences

Hard to develop content

Device specific solutions

Limited free content

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

01 **Enable interactive and collaborative mechanisms** for AR solutions

02 Allow **interoperability** across different platforms

03 Discover **limitations of current AR applications** for education

04 **Identify the requirements** to simplify adoption of AR in the classroom

05 **Develop and evaluate** working solutions

ARETE supports the interactive technologies efforts, through the multi-user interactions within AR technologies evaluated in education





- ▶ The main objective of this work is to explore and propose innovative solutions which use AR in the educational sector, with a strong focus on collaborative, multi-user interactions



- ▶ The main objective of this work is to explore and propose innovative solutions which use AR in the educational sector, with a strong focus on collaborative, multi-user interactions



- ▶ Define a software architecture that simplifies the creation and adoption of collaborative AR solutions for education

10

OBJECTIVES

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

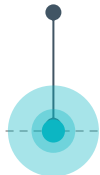
Future work

Main objective



Objective 1

Identify the causes behind the lack of adoption of AR solutions in schools.

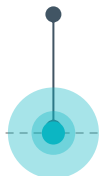


Main objective



Objective 1

Identify the causes behind the lack of adoption of AR solutions in schools.



Objective 2

Develop a software library enabling the creation of multi-user AR applications.

Main objective



Objective 1

Identify the causes behind the lack of adoption of AR solutions in schools.

Objective 3

Define a modular architecture that fulfils the requirements and design objectives for AR in education.

Objective 2

Develop a software library enabling the creation of multi-user AR applications.

Main objective



Objective 1

Identify the causes behind the lack of adoption of AR solutions in schools.

Objective 3

Define a modular architecture that fulfils the requirements and design objectives for AR in education.

Objective 2

Develop a software library enabling the creation of multi-user AR applications.

Objective 4

Validate the architecture by creating and testing a collaborative multi-platform AR application.

Main objective



INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work



INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

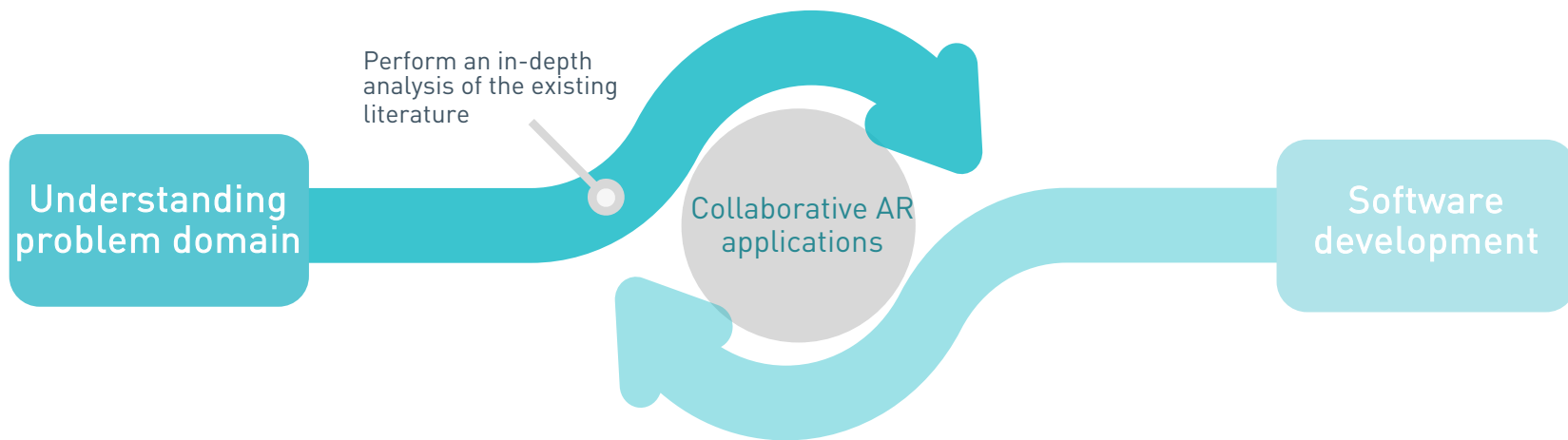
RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work



INTRODUCTION

Motivation
Objectives

Methodology

Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

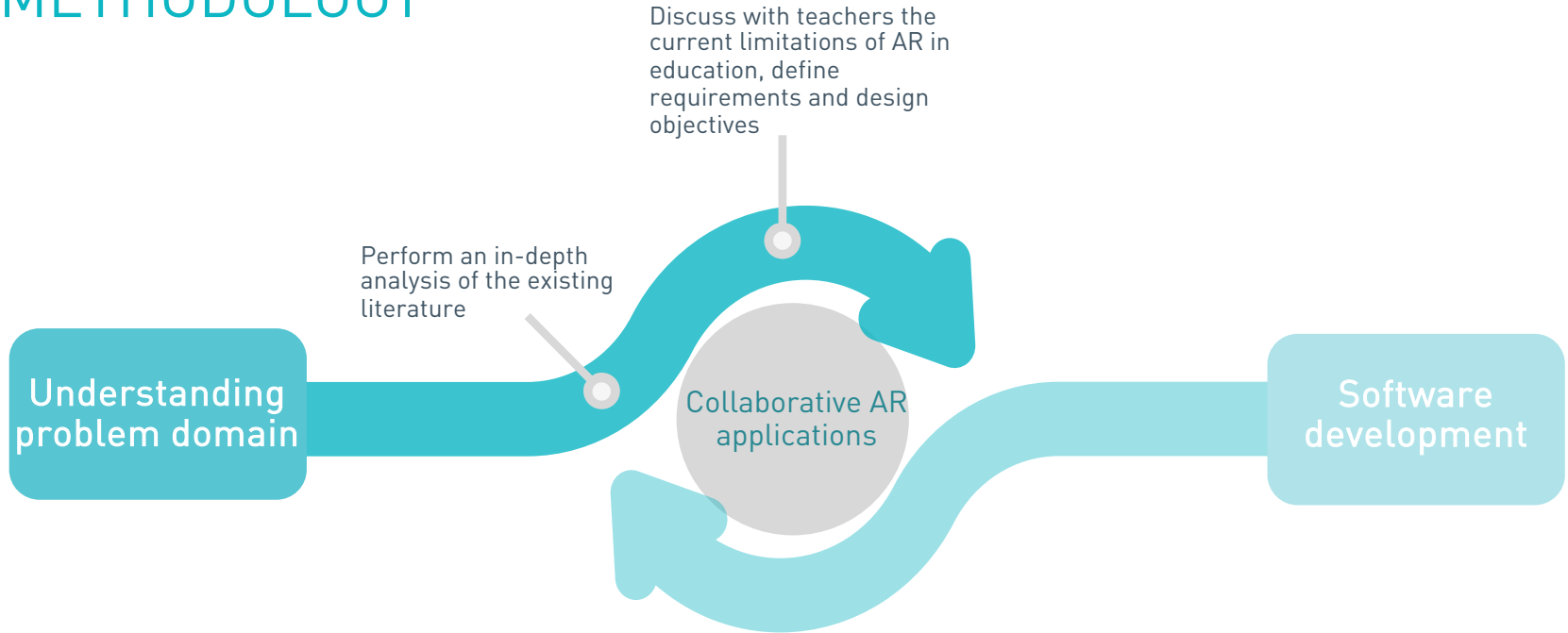
RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work



INTRODUCTION

Motivation
Objectives

Methodology

Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

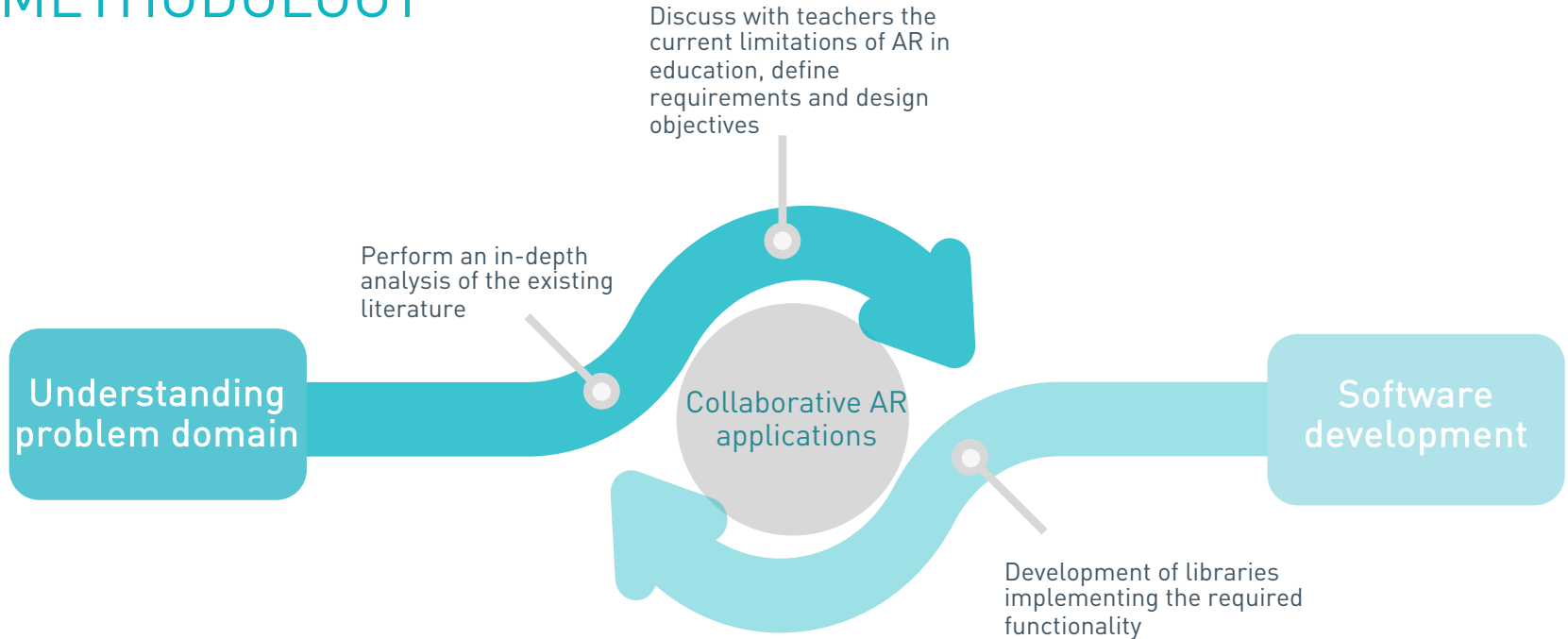
RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work



INTRODUCTION

Motivation
Objectives

Methodology

Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

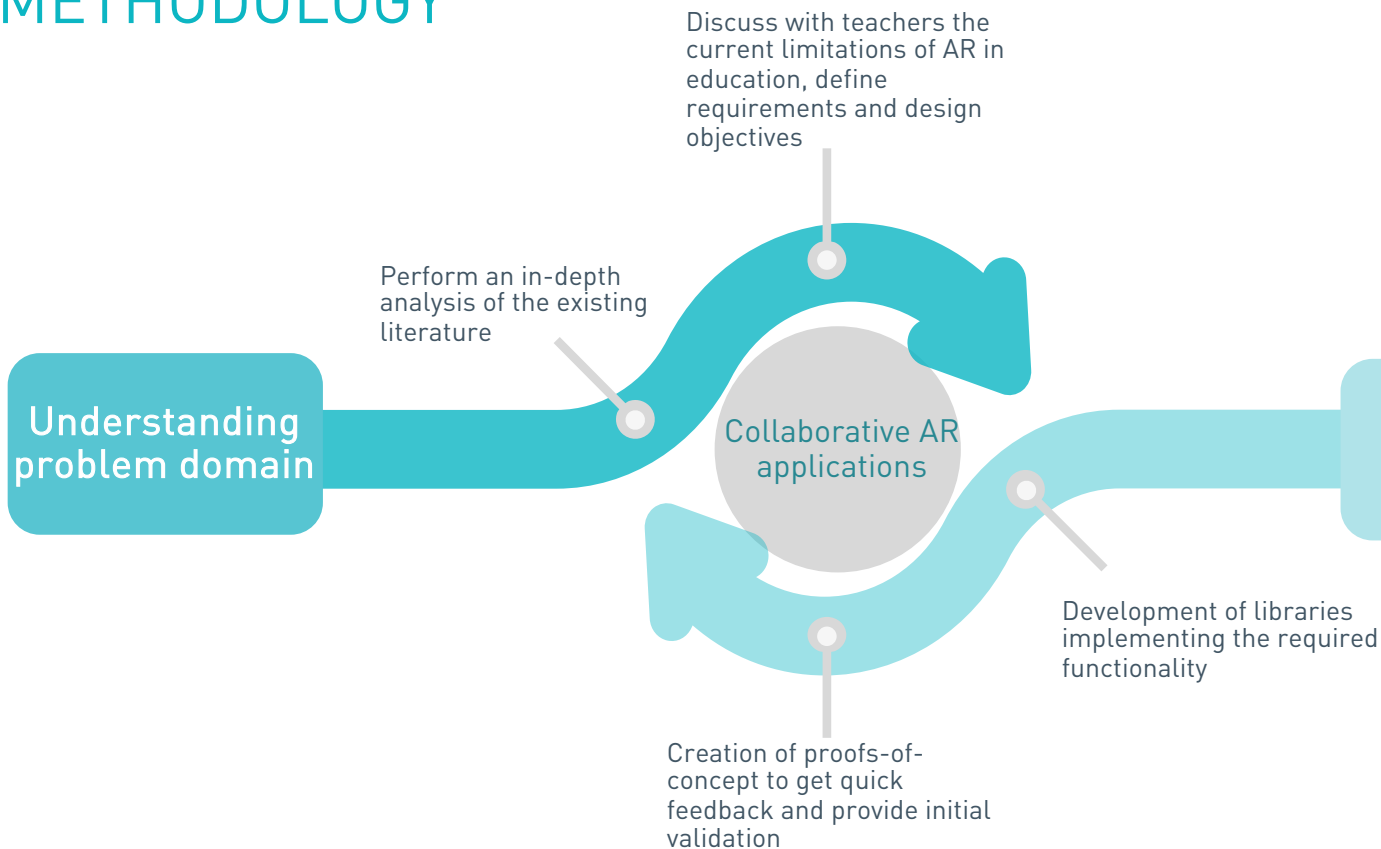
RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work



12

CONTRIBUTIONS

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

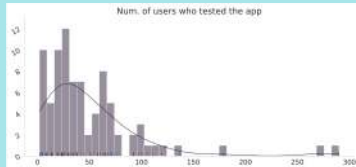
CONCLUSIONS

Conclusions

Publications

Future work

C1

In-depth analysis of the
state-of-the-art

- State of research on AR applied to education
- Motivation for using AR
- Effects of AR on motivation and retention

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

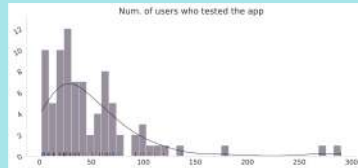
CONCLUSIONS

Conclusions

Publications

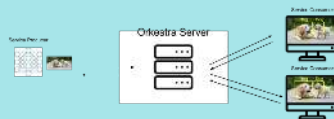
Future work

C1

In-depth analysis of the
state-of-the-art

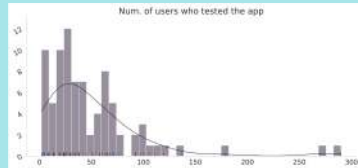
- State of research on AR applied to education
- Motivation for using AR
- Effects of AR on motivation and retention

C2

Orkestra: library for
multi-user interactions

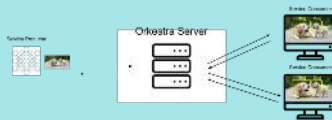
- Low-latency
- Media synchronization
- Multiplatform

C1

In-depth analysis of the
state-of-the-art

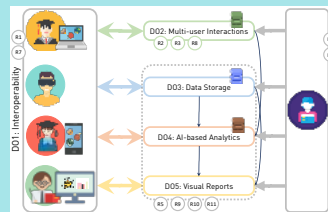
- State of research on AR applied to education
- Motivation for using AR
- Effects of AR on motivation and retention

C2

Orkestra: library for
multi-user interactions

- Low-latency
- Media synchronization
- Multiplatform

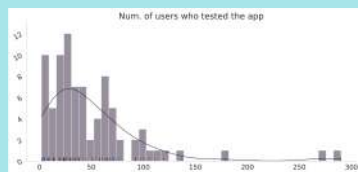
C3

cleAR: modular
architecture for AR
solutions

- Satisfies teacher requirements
- Simplifies creation of collaborative applications

C1

In-depth analysis of the
state-of-the-art



- State of research on AR applied to education
- Motivation for using AR
- Effects of AR on motivation and retention

C2

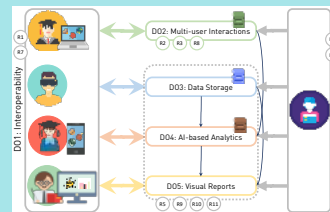
Orkestra: library for
multi-user interactions



- Low-latency
- Media synchronization
- Multiplatform

C3

cleAR: modular
architecture for AR
solutions



- Satisfies teacher requirements
- Simplifies creation of collaborative applications

C4

ARoundTheWorld:
implementation of an AR
app based on cleAR



- Tested in 3 educational institutions
- Validates cleAR design objectives

2. STATE OF THE ART

Related Work

Literature Review

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative
AR in education

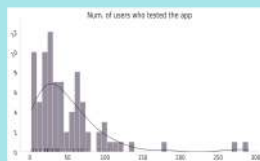
Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

C1

In-depth analysis of
the state-of-the-art



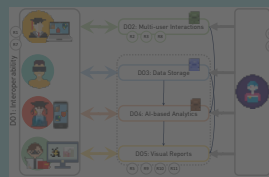
C2

Orkestra: library for
multi-user
interactions



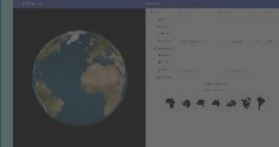
C3

cleAR: modular
architecture for AR
solutions



C4

ARoundTheWorld:
implementation of an
AR app based on
cleAR



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Steady year-to-year increase of publications

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview

Methodology
Results
Discussion

RESEARCH
RESULTS

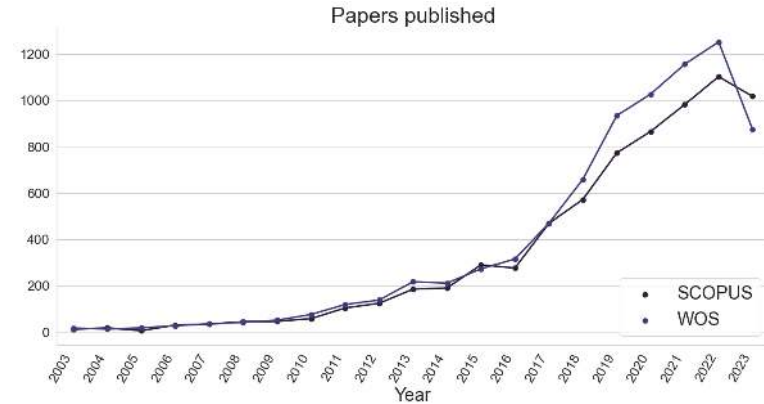
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- Steady year-to-year increase of publications



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

RESEARCH RESULTS

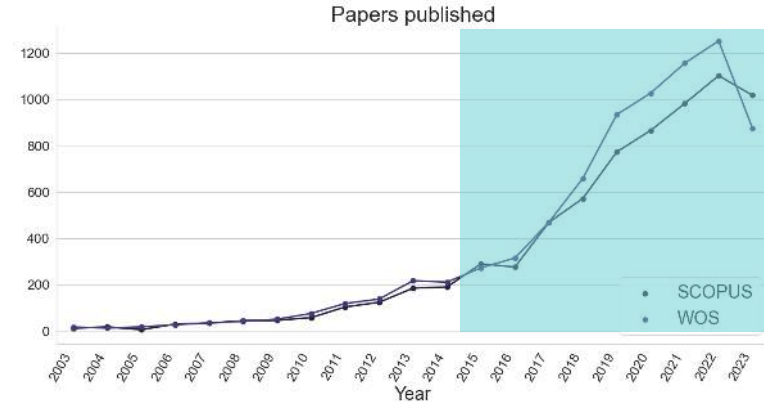
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- Steady year-to-year increase of publications



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview

Methodology
Results
Discussion

RESEARCH RESULTS

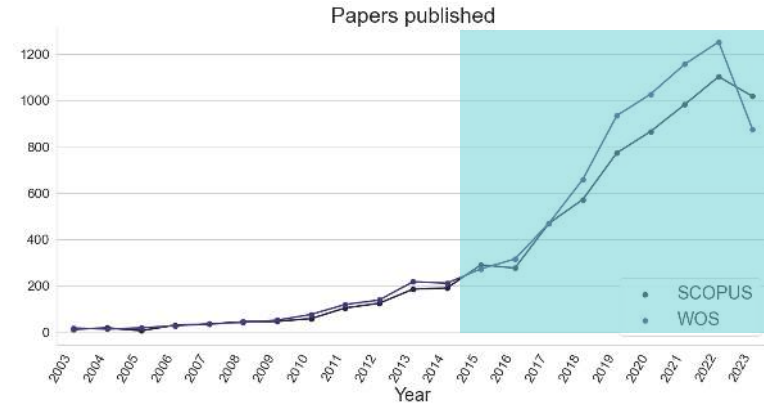
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Steady year-to-year increase of publications
- ▶ Several systematic reviews covering different aspects of AR in education



INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

CONCLUSIONS

Conclusions

Publications

Future work

- ▶ Steady year-to-year increase of publications
- ▶ Several systematic reviews covering different aspects of AR in education

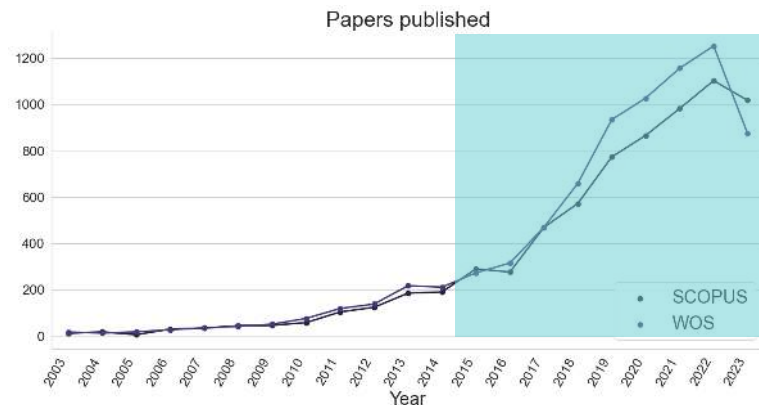
Advantages
and
drawbacks

Status of the
technology

AR for STEM

AR for
language
learning

Evaluation of
usage in
schools



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview

Methodology
Results
Discussion

RESEARCH RESULTS

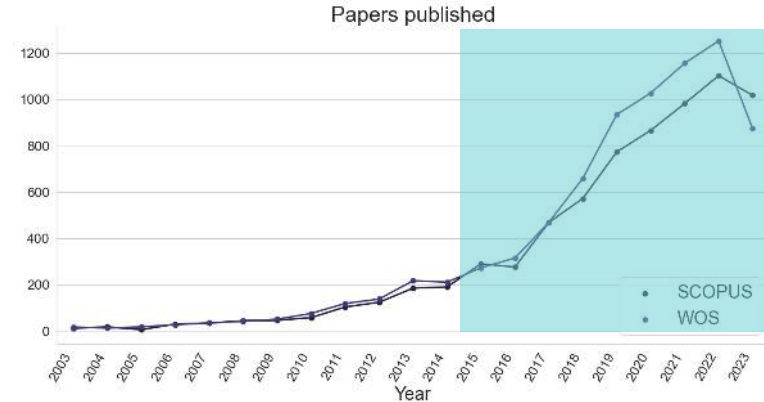
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Steady year-to-year increase of publications
- ▶ Several systematic reviews covering different aspects of AR in education



Advantages
and
drawbacks

Status of the
technology

AR for STEM

AR for
language
learning

Evaluation of
usage in
schools

AR-based
serious
games

Impact in
learning
effectiveness

Usability and
UX

Bibliometric
analysis

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview

Methodology
Results
Discussion

RESEARCH RESULTS

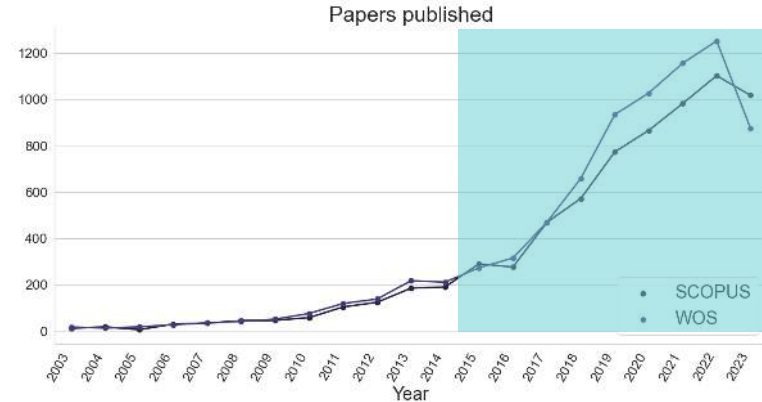
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Steady year-to-year increase of publications
- ▶ Several systematic reviews covering different aspects of AR in education



Advantages
and
drawbacks

Status of the
technology

AR for STEM

AR for
language
learning

Evaluation of
usage in
schools

AR-based
serious
games

Impact in
learning
effectiveness

Usability and
UX

Bibliometric
analysis

Collaborative
AR
(2014, 8 papers)

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

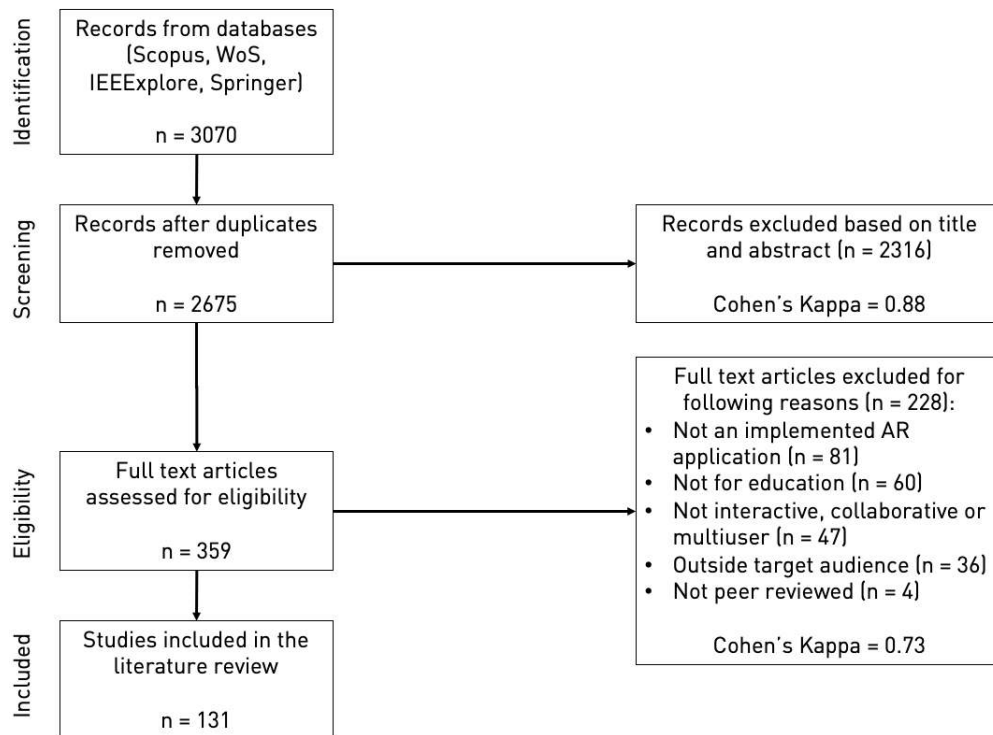
Future work

- ▶ Systematic literature review of AR applications for education

RQ1 What collaborative, multi-user, interactive AR applications in education?

RQ2 What is the motivation for using AR apps as an educational tool?

RQ3 Are AR applications effective at improving the knowledge of the students?



- ▶ Inclusion criteria:
 - ▶ Studies published between 2015 and 2023
 - ▶ New AR application
 - ▶ Primary and secondary schools

- ▶ Exclusion criteria:
 - ▶ Not interactive, collaborative or multi-user
 - ▶ Not for education
 - ▶ Not peer reviewed

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

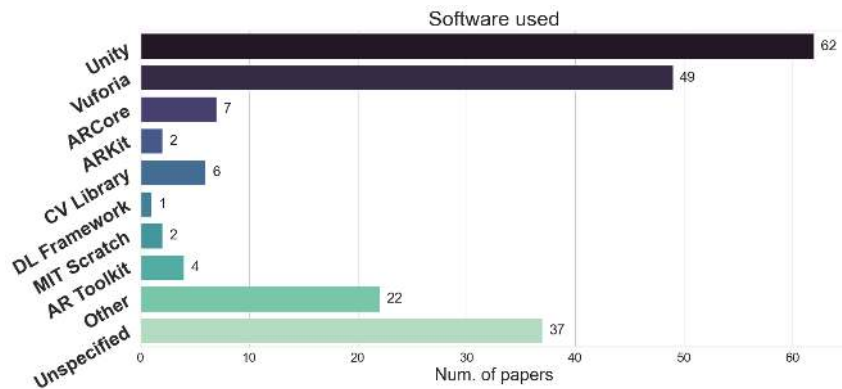
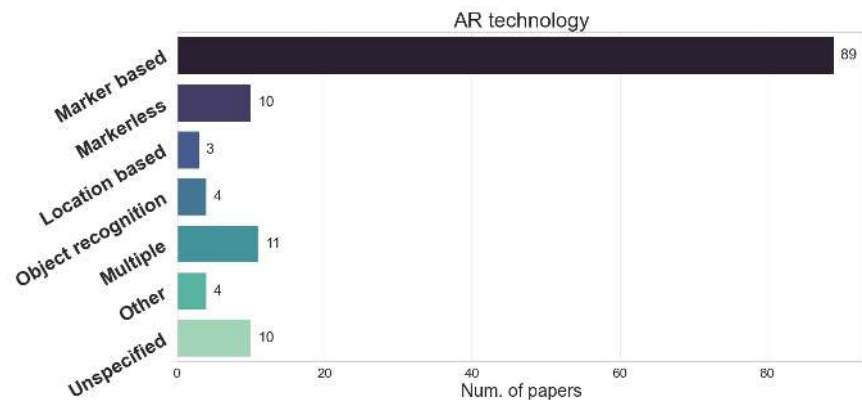
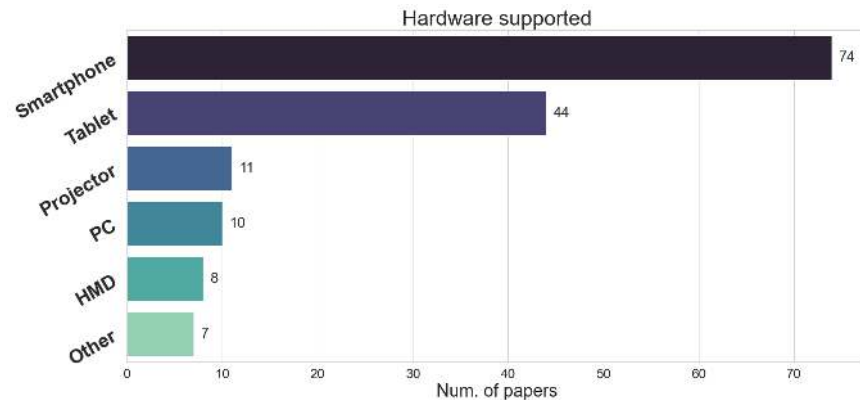
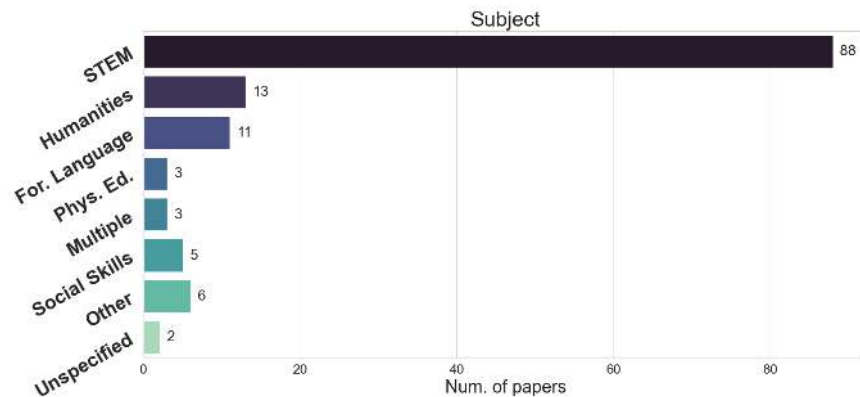
Validation of the proposed architecture

CONCLUSIONS

Conclusions

Publications

Future work



RQ1: Interactive, multi-user, collaborative AR in education

01

Studies were categorized based on their **interactive functionalities**:

Basic interactivity / Object interaction / Quiz / Behaviour tracking / Augmented interactions / **Multi-user** / **Collaborative**

02

Majority of the studies offer **very limited** interactive functionalities

03

Mostly **single-user experiences** (21/131 multi-user, 16/131 collaborative)

04

Behaviour tracking hardly considered (6/131)

05

Augmented interactions started getting traction in 2019

RQ2: Motivation for AR as educational tool

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

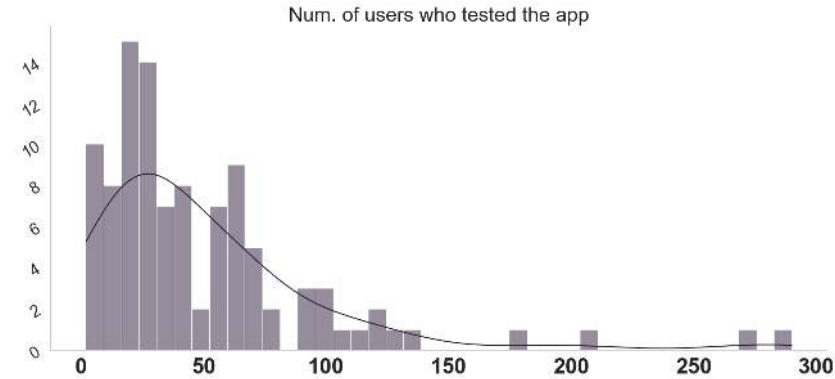
CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Most of the studies **do not motivate** the choice of their application
- ▶ They usually present the advantages provided by using AR in the classroom:
 - ▶ Interaction with real world
 - ▶ Simplifies gamification
 - ▶ It can be used as a tool to provide supplementary material
 - ▶ Enables 3D visualization and animations
 - ▶ Can be especially helpful for children with learning disabilities
 - ▶ (Limited) effect on student motivation

RQ3: Effectiveness of AR at improving knowledge

- ▶ 103 out of 131 studies provided information on the number of participants



INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

CONCLUSIONS

Conclusions

Publications

Future work

RQ3: Effectiveness of AR at improving knowledge

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

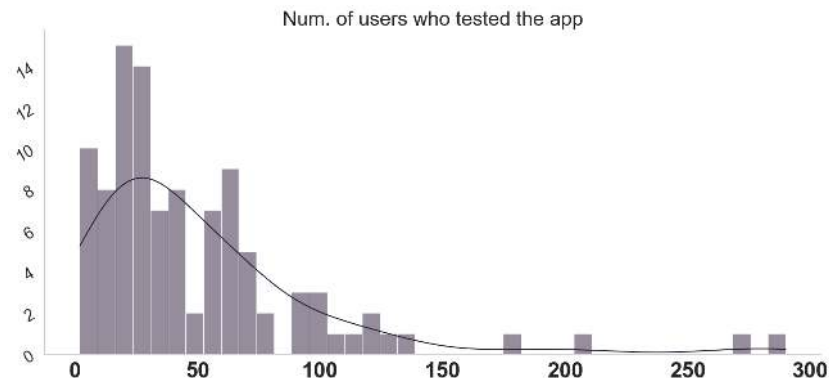
CONCLUSIONS

Conclusions

Publications

Future work

- ▶ 103 out of 131 studies provided information on the number of participants
- ▶ Evaluation performed in three ways:
 - ▶ Comparison with control group
 - ▶ Surveys
 - ▶ Pre- and post-test



RQ3: Effectiveness of AR at improving knowledge

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

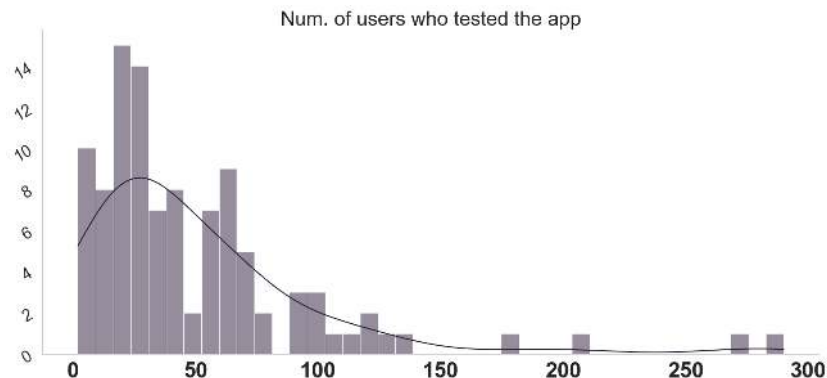
CONCLUSIONS

Conclusions

Publications

Future work

- ▶ 103 out of 131 studies provided information on the number of participants
- ▶ Evaluation performed in three ways:
 - ▶ Comparison with control group
 - ▶ Surveys
 - ▶ Pre- and post-test
- ▶ Studies rarely present statistically significant results



RQ3: Effectiveness of AR at improving knowledge

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

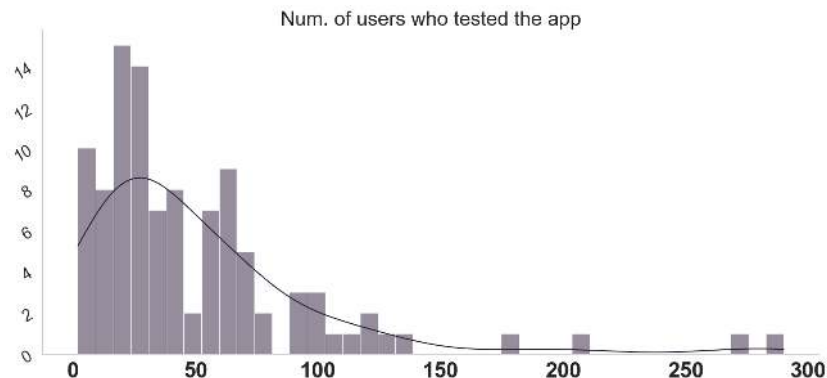
CONCLUSIONS

Conclusions

Publications

Future work

- ▶ 103 out of 131 studies provided information on the number of participants
- ▶ Evaluation performed in three ways:
 - ▶ Comparison with control group
 - ▶ Surveys
 - ▶ Pre- and post-test
- ▶ Studies rarely present statistically significant results
- ▶ Limited improvement over traditional learning methods



RQ3: Effectiveness of AR at improving knowledge

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology

Results

Discussion

RESEARCH RESULTS

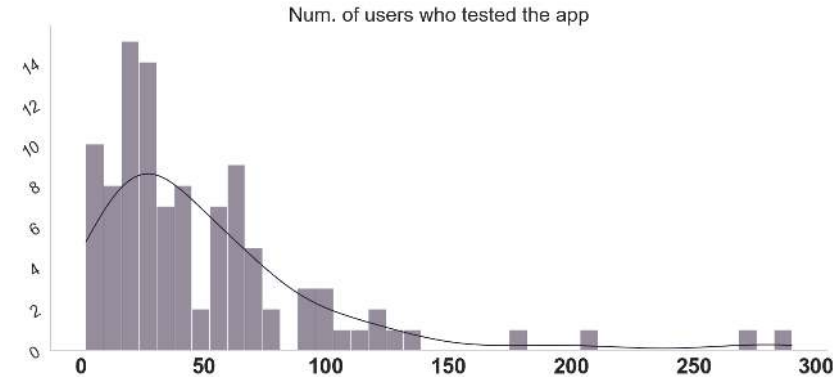
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ 103 out of 131 studies provided information on the number of participants
- ▶ Evaluation performed in three ways:
 - ▶ Comparison with control group
 - ▶ Surveys
 - ▶ Pre- and post-test
- ▶ Studies rarely present statistically significant results
- ▶ Limited improvement over traditional learning methods
- ▶ No studies showing negative impacts of AR – only 1 showing no effect



RQ3: Effectiveness of AR at improving knowledge

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

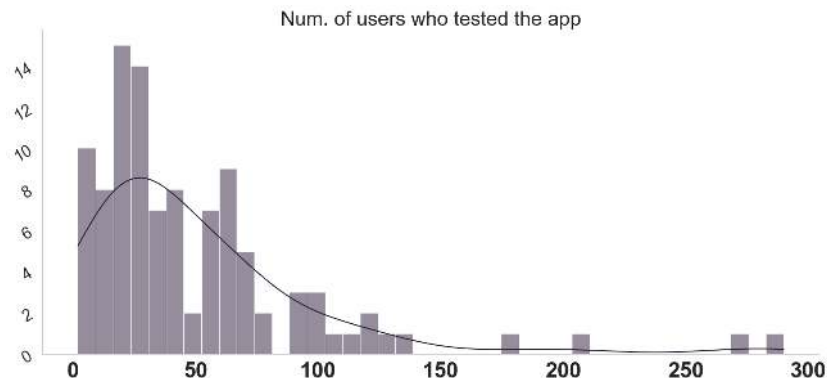
CONCLUSIONS

Conclusions

Publications

Future work

- ▶ 103 out of 131 studies provided information on the number of participants
- ▶ Evaluation performed in three ways:
 - ▶ Comparison with control group
 - ▶ Surveys
 - ▶ Pre- and post-test
- ▶ Studies rarely present statistically significant results
- ▶ Limited improvement over traditional learning methods
- ▶ No studies showing negative impacts of AR – only 1 showing no effect
- ▶ Only 2 studies measuring topics retention



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results

Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

01

Issues

Lack of authoring tools

Availability of free-to-use 3D content

Code publication

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

01

Issues

Lack of authoring tools

Availability of free-to-use 3D content

Code publication

02

Stronger positive impact is related to **inclusion of AR usage in existing school curricula**

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

01

Issues

Lack of authoring tools

Availability of free-to-use 3D content

Code publication

02

Stronger positive impact is related to **inclusion of AR usage in existing school curricula**

03

Multi-user AR for distance learning is an **unexplored topic**

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

01

Issues

Lack of authoring tools

Availability of free-to-use 3D content

Code publication

02

Stronger positive impact is related to **inclusion of AR usage in existing school curricula**

03

Multi-user AR for distance learning is an **unexplored topic**

04

Latest AR technologies **still not widely used** (mostly vision, QR-based)

3. RESEARCH RESULTS

Main contributions of the PhD work

Interoperable architecture for collaborative AR in education

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

CONCLUSIONS

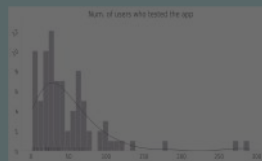
Conclusions

Publications

Future work

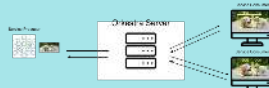
C1

In-depth analysis of the state-of-the-art



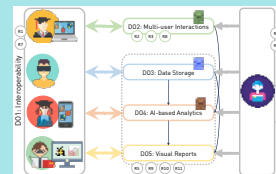
C2

Orchestra: library for multi-user interactions



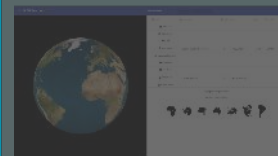
C3

cleAR: modular architecture for AR solutions



C4

ARoundTheWorld: implementation of an AR app based on cleAR



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

Mostly single user apps

No integration with
school curricula

No data collection

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

Mostly single user apps

No integration with
school curricula

No data collection

**TEACHERS INPUT**

How AR can
positively impact
education

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

Mostly single user apps

No integration with
school curricula

No data collection



TEACHERS INPUT

How AR can
positively impact
education

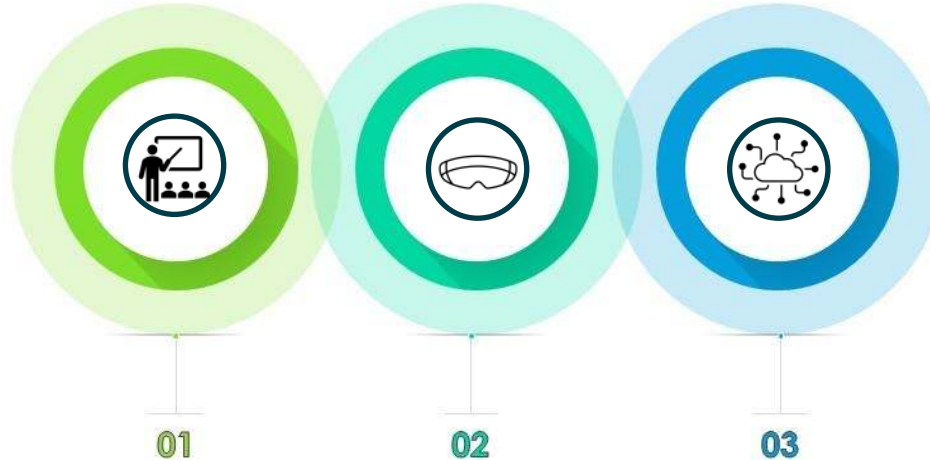


FRAMEWORK

Create AR content
satisfying
requirements

TEACHER SURVEY – In cooperation with Ikastolen Elkartea

- ▶ 45 questions split into 3 sections:



Information about the teachers and **usage of TEL** at their school

Usage of AR apps in school and perception of AR as teaching instrument

Which **technological tools** would they like to use at school

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

01

47 teachers participated –
mostly **facilitators**

02

High availability of **PCs
and laptops**

03

Widespread adoption of
software tools

04

Dissatisfaction with
existing AR applications

05

AR should provide
interactivity & collaboration

06

Interest in using **Artificial
Intelligence** tools

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

RESEARCH RESULTS

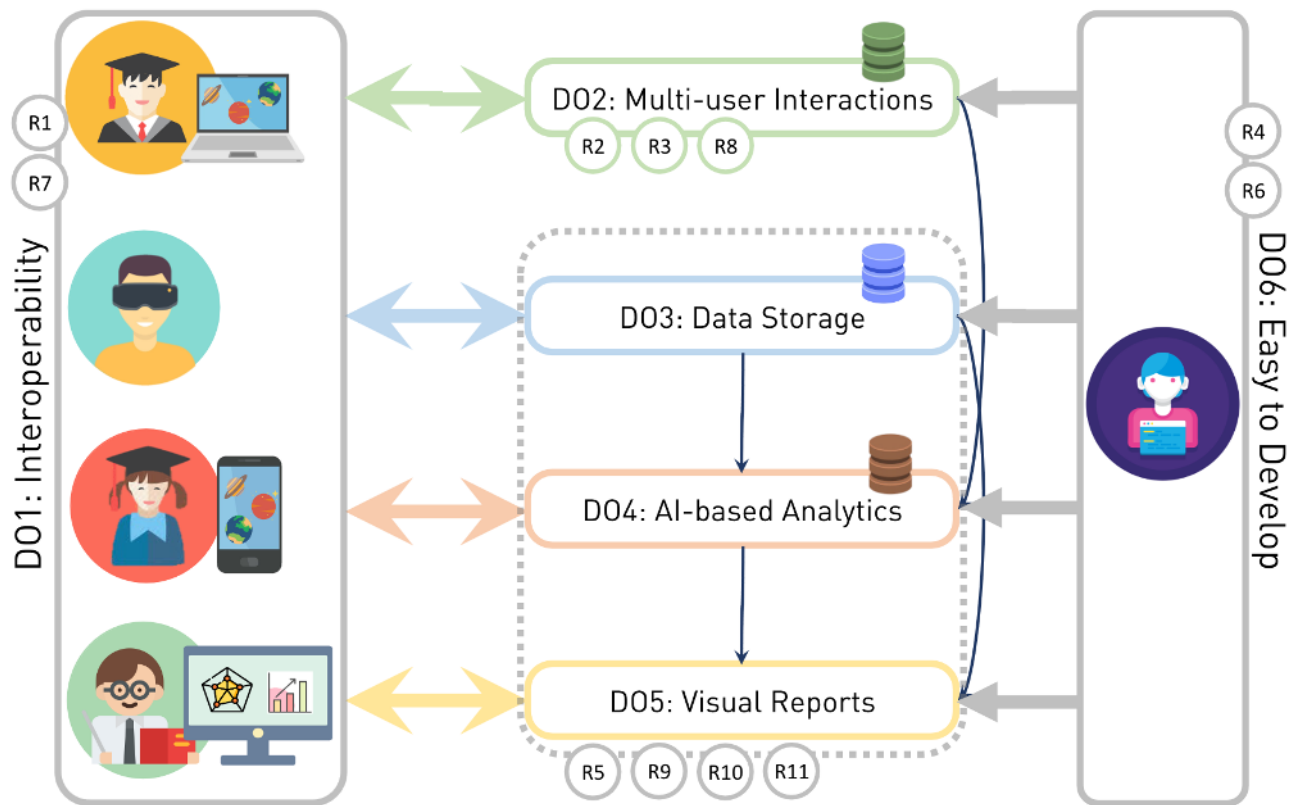
Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

R1	Work on many different platform and devices
R2	Foster teacher-student interactivity
R3	Foster student-student interactivity
R4	Enable several customisation options
R5	Collect data on how the application is used
R6	Enable easy development, as well as installing and updating
R7	Work smoothly on older hardware
R8	Allow collaborative work
R9	Provide tools for analysis of data and usage patterns
R10	Enable automatic analysis of test difficulty
R11	Detect students with learning difficulties



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

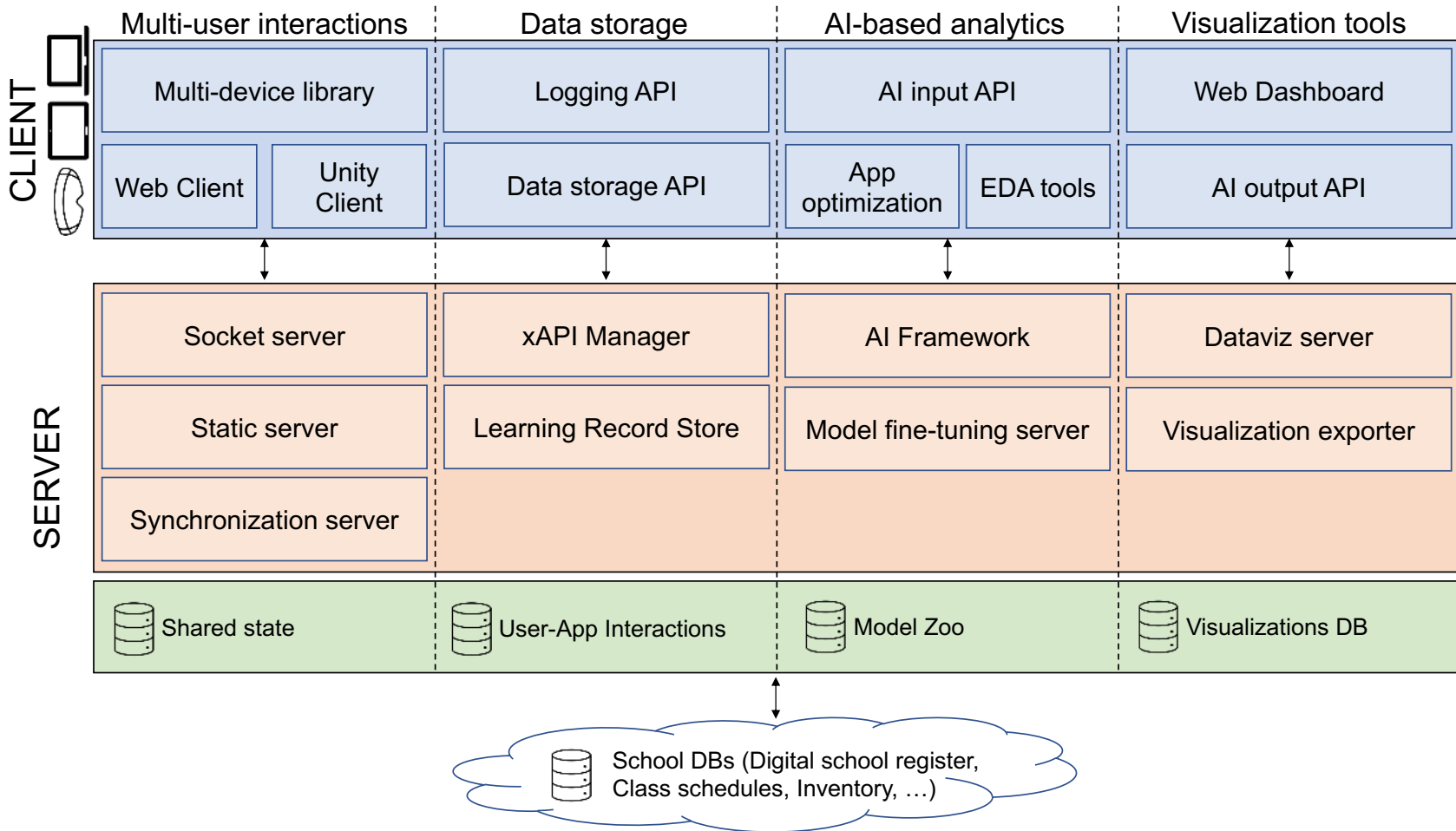
Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

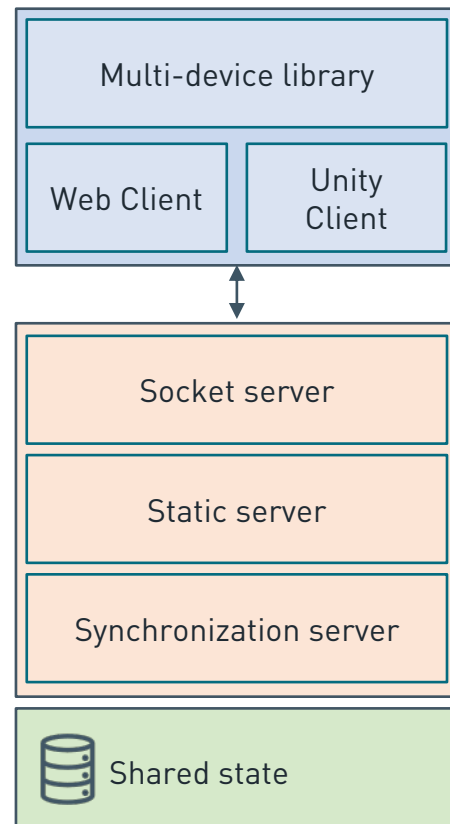
Conclusions
Publications
Future work

- ▶ *cleAR* is the architecture defined to fulfil every design objective
- ▶ It is a client-server architecture, composed by several modules
- ▶ Each module oversees a specific task and satisfies a design objectives
- ▶ D01 (interoperability) and D06 (ease of development) fulfilled thanks to how the architecture was defined
- ▶ Some components rely on existing libraries, developed at Vicomtech or available as open source

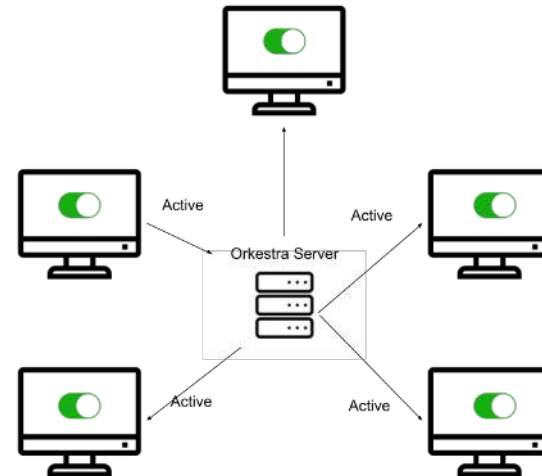


- ▶ Synchronisation and management of heterogeneous devices in multi-user applications
- ▶ Developed for web applications, then extended to other software environments
- ▶ Enables context maintenance in multi-device environments
- ▶ Main features:
 - ▶ State sharing and synchronisation
 - ▶ Service publishing from one device to another
 - ▶ Video synchronisation from a centralised clock

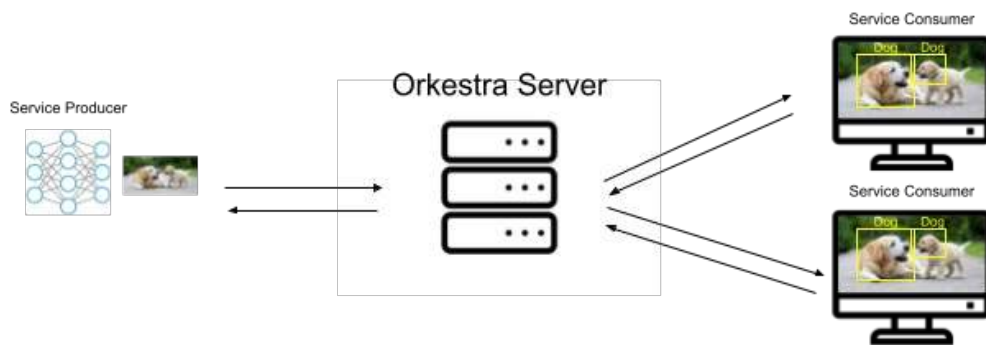
Multi-user interactions



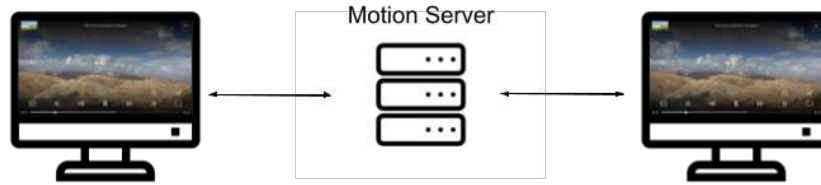
- ▶ Applications can store data and share it across devices
- ▶ When a device changes its state, this is shared across every connected device
- ▶ There are two types of state:
 - ▶ Agent
 - ▶ Application



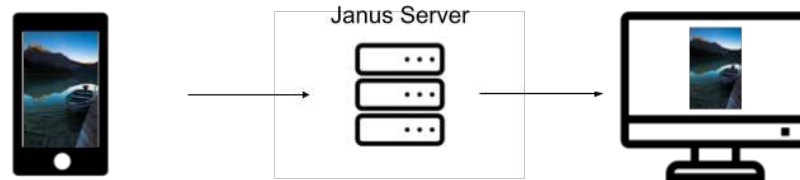
- ▶ Create services on a device which are then consumed by others
- ▶ Resource intensive services can be delegated to powerful devices
- ▶ Example: computer vision tasks



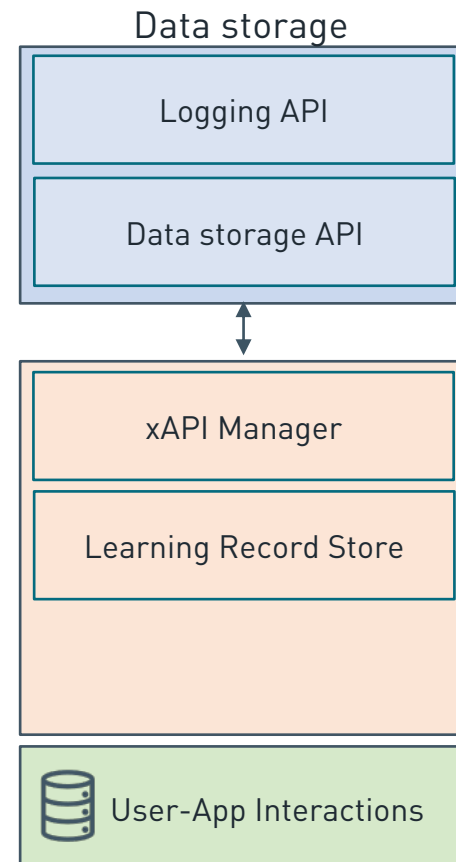
- ▶ Multiple user can watch the same video content at the same time



- ▶ Multimedia content can also be shared across devices, for example to share the screen or the content grabbed from a camera



- ▶ Data can be serialized and stored on a DB or locally
- ▶ An API allows storing and retrieving data through a Learning Record Store
- ▶ This enables applications to fetch data from existing LMSs (*e.g.*, a set of questions for a test) and writing data to it (*e.g.*, results of in-app quizzes, students' progress)



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

RESEARCH RESULTS

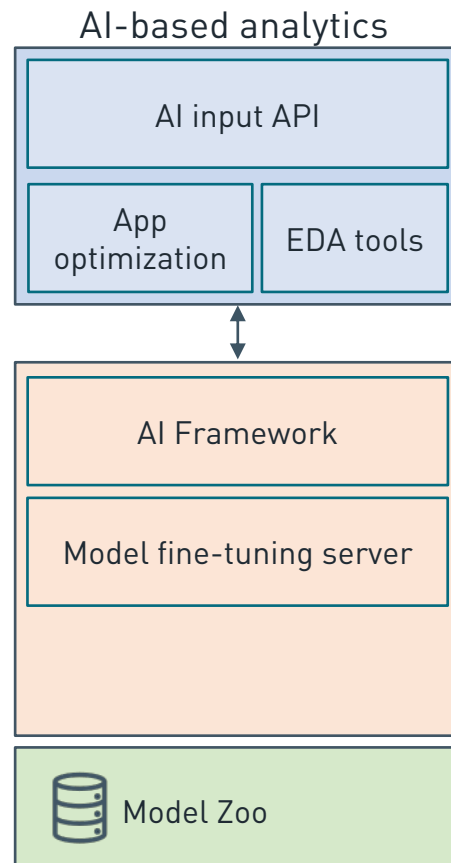
Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Data collected through the application can be used to train or fine-tune AI models
- ▶ Models are trained and stored on the server side
- ▶ The API defines an interface for data input, and relies on existing AI frameworks
- ▶ Specific algorithms for optimization of app parameters



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

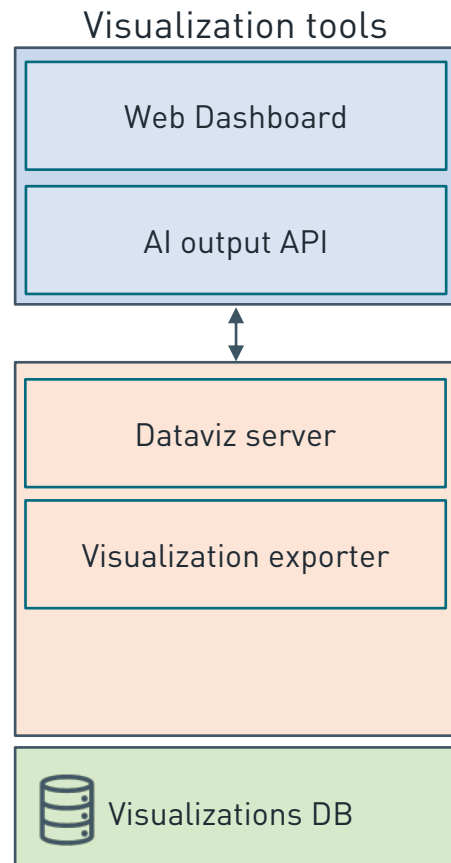
Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Provides an interface for visualisation and reporting
- ▶ Simplifies the exploration of the data collected
- ▶ Low-code creation of plots and charts
- ▶ Enables teachers to draw insights



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Three PoCs were created to perform a technical validation of the architecture and to verify that different design objectives were satisfied

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative AR
in education

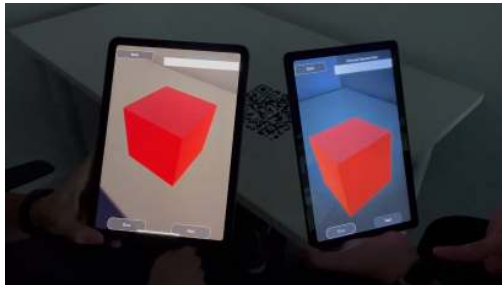
Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

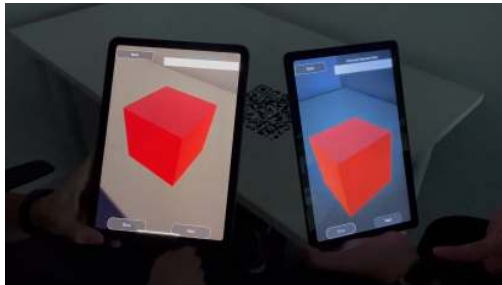
- ▶ Three PoCs were created to perform a technical validation of the architecture and to verify that different design objectives were satisfied

AR Cube



- ▶ Three PoCs were created to perform a technical validation of the architecture and to verify that different design objectives were satisfied

AR Cube

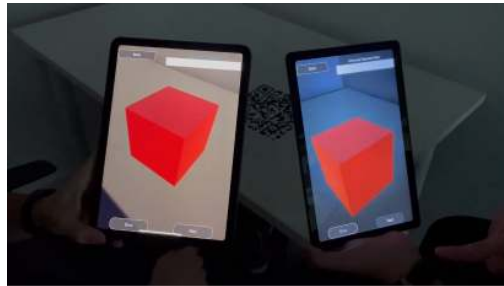


xAPI data analysis

A screenshot of a web application interface for xAPI data analysis. At the top, there is a text input field labeled "Enter some text" and a "Send" button. Below this, there are three blue buttons stacked vertically: "Send 1 statement", "Send 5 statements", and "Send 20 statements". At the bottom of the interface, there are three icons with labels: a bar chart icon labeled "Access Learning, Lecture", a circular logo icon labeled "Access Learning, Lecturebooks", and a clock icon labeled "Start stress test".

- ▶ Three PoCs were created to perform a technical validation of the architecture and to verify that different design objectives were satisfied

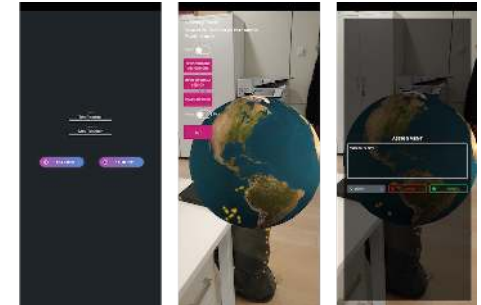
AR Cube



xAPI data analysis

 A screenshot of a web application for xAPI data analysis. It features a text input field labeled "Enter some text" with a "Send" button. Below the input are three buttons: "Send 1 statement", "Send 5 statements", and "Send 20 statements". At the bottom, there are three icons with labels: a bar chart for "Access Learning, Lecture", a circular logo for "Access Learning, Lecture", and a clock for "Start stress test".

AR Geo Quiz



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Interactions with the augmented content are broadcast to all connected users
- ▶ Users can rotate the object or change its colour
- ▶ Interactions are stored in a queue and sent when the *dispatch time* has elapsed
- ▶ *Dispatch time* can also be estimated automatically

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

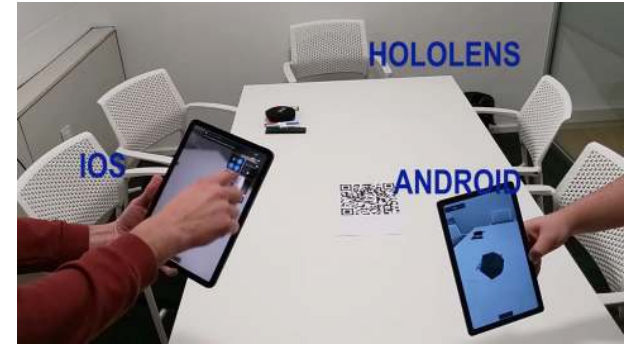
Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Interactions with the augmented content are broadcast to all connected users
- ▶ Users can rotate the object or change its colour
- ▶ Interactions are stored in a queue and sent when the *dispatch time* has elapsed
- ▶ *Dispatch time* can also be estimated automatically



INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative AR
in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

- ▶ A Learning Record Store that collects data from users accessing a web page.
- ▶ Data collected in the form of xAPI statements
- ▶ Demonstrates integration with Learning Locker
- ▶ Stress test was performed with 10 clients, sending a total of 80k statements
- ▶ The statements were used to train a simple linear classifier

 [Access Learning Locker](#)

 [Access Jupyter Notebooks](#)

 [Start stress test](#)

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative AR
in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ A more complex example that simulates a geography quiz
- ▶ This PoC was then further developed to create *ARoundTheWorld*
- ▶ Two modalities:
 - ▶ *free exploration*
 - ▶ *forced perspective*



INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative AR
in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

	Design Objective				
PoC	Interoperable	Multi-user	Data Storage	AI Support	Easy to develop
AR Cube	✓	✓		✓	✓
xAPI Data Analysis		✓	✓	✓	✓
AR Geography Quiz	✓	✓	✓		

Validation of the proposed architecture

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

CONCLUSIONS

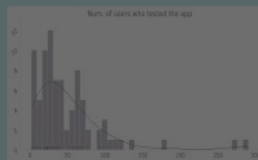
Conclusions

Publications

Future work

C1

In-depth analysis of the state-of-the-art



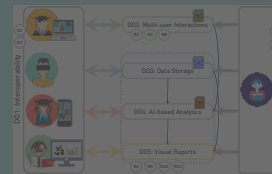
C2

Orchestra: library for multi-user interactions



C3

cleAR: modular architecture for AR solutions



C4

ARoundTheWorld: implementation of an AR app based on cleAR



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

01

Conceptual validation of cleAR performed through the creation of 3 PoCs

02

A more complex application is required to **validate cleAR in a school environment**

03

The app, *ARoundTheWorld*, was **designed with the help of teachers**

04

Objective: demonstrate potential of *cleAR* in the creation of **collaborative and interoperable applications**

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

01

Collaborative geography quiz

02

App sends a question, student answers by **placing pin on 3D globe**

03

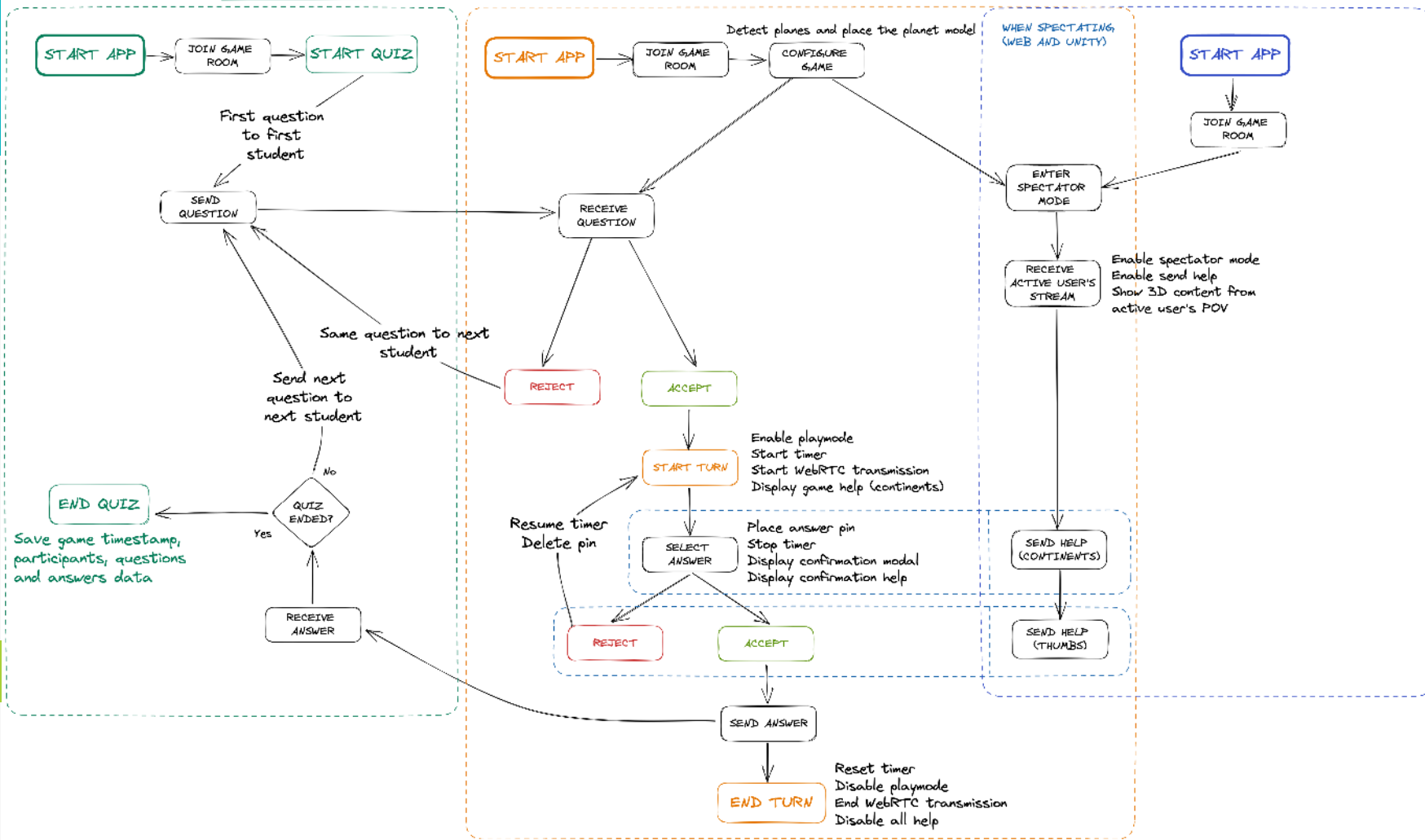
Other students **collaborate by giving hints**

04

Teacher has access to **dashboard**, to **display results** or **create new quiz**

05

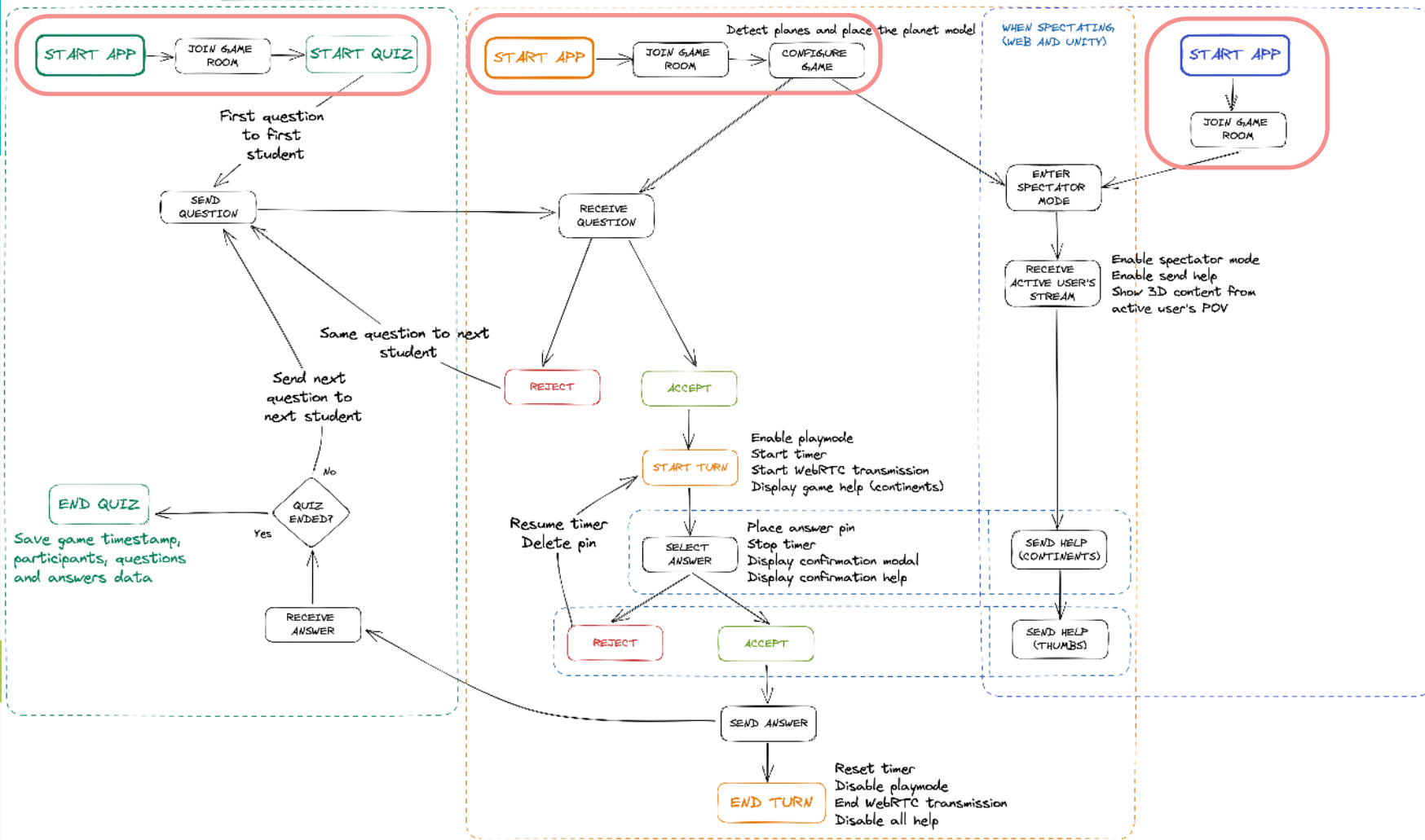
App is **gamified** to motivate students

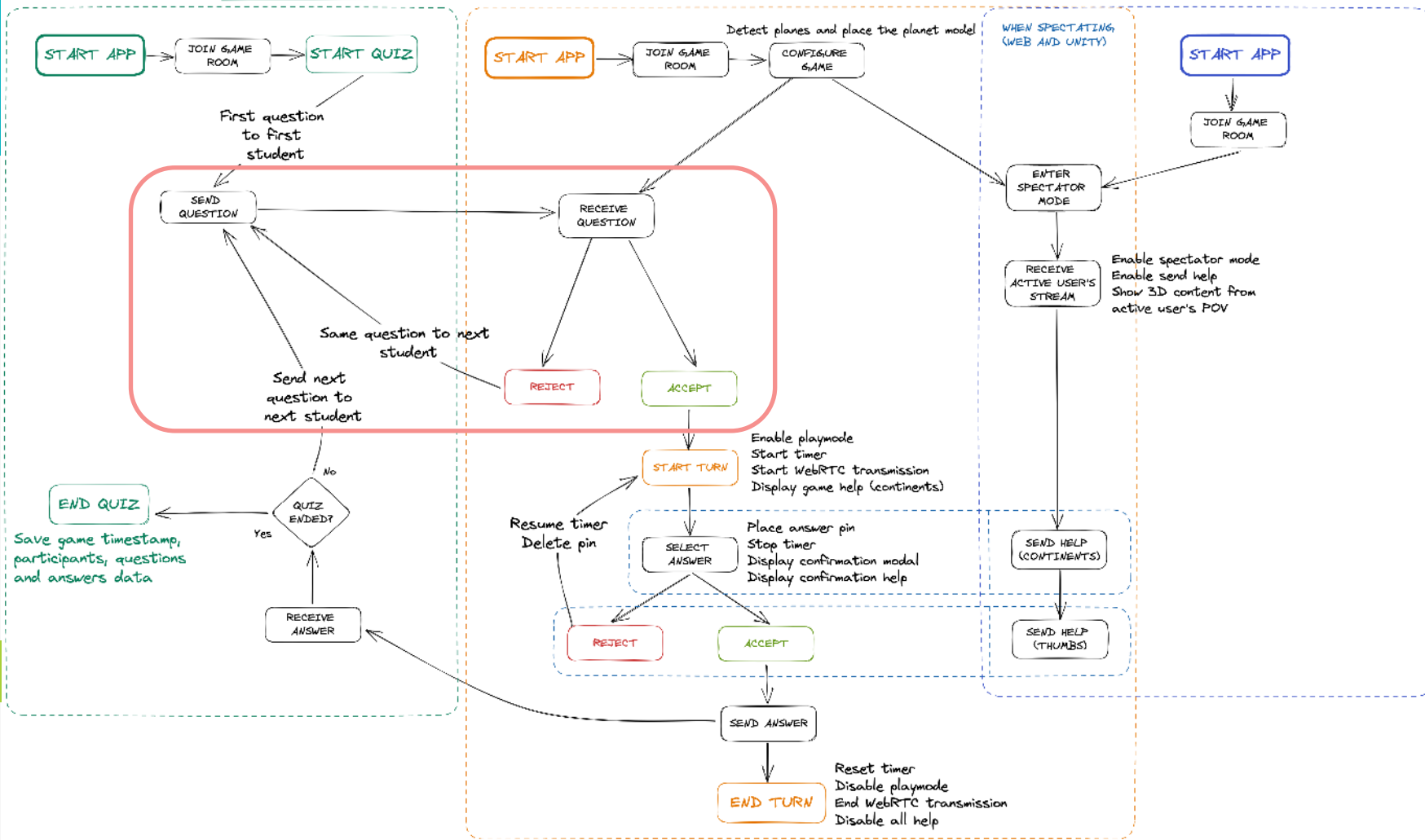
WEB APP (TEACHER)UNITY APP (STUDENT)WEB APP (STUDENT)

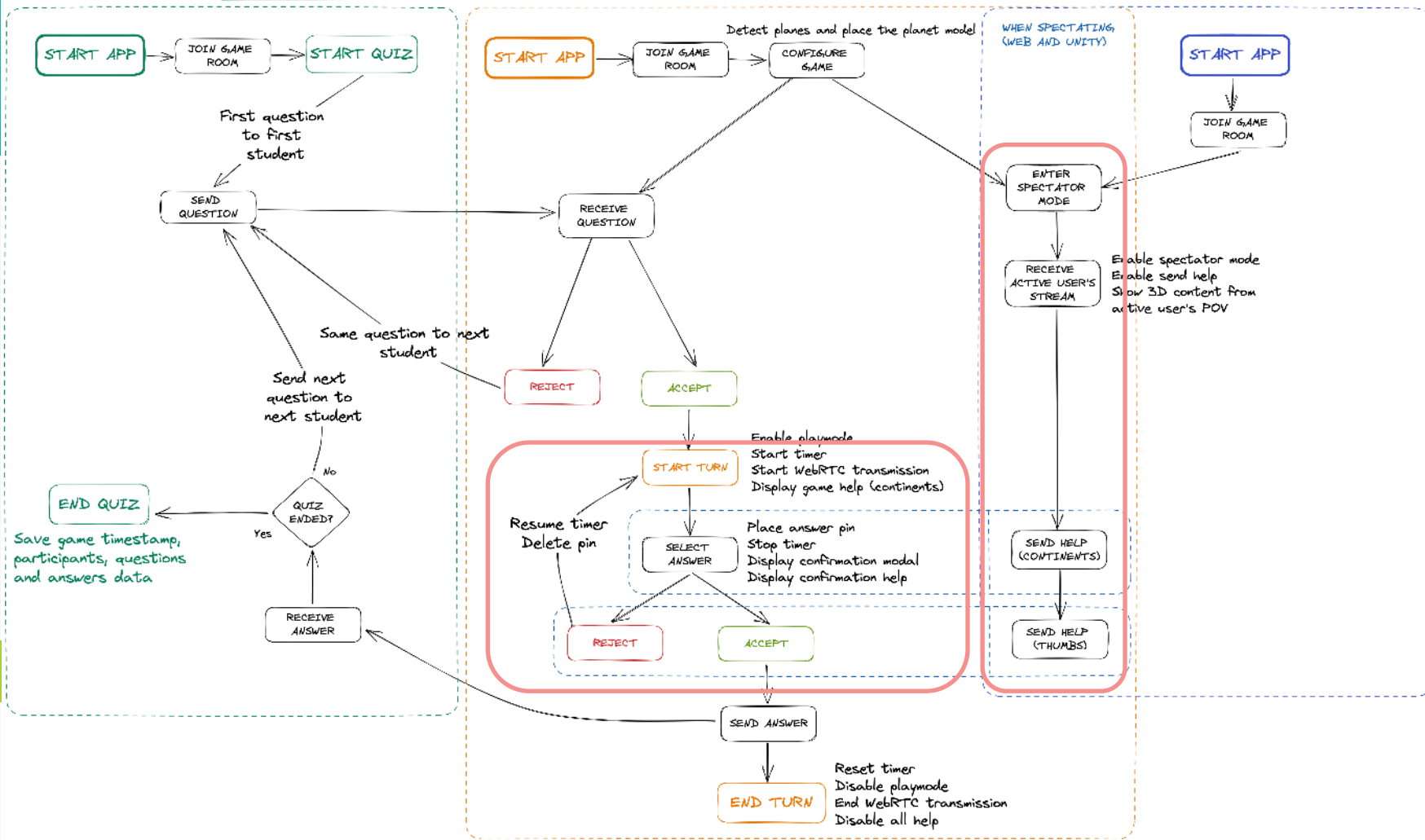
WEB APP (TEACHER)

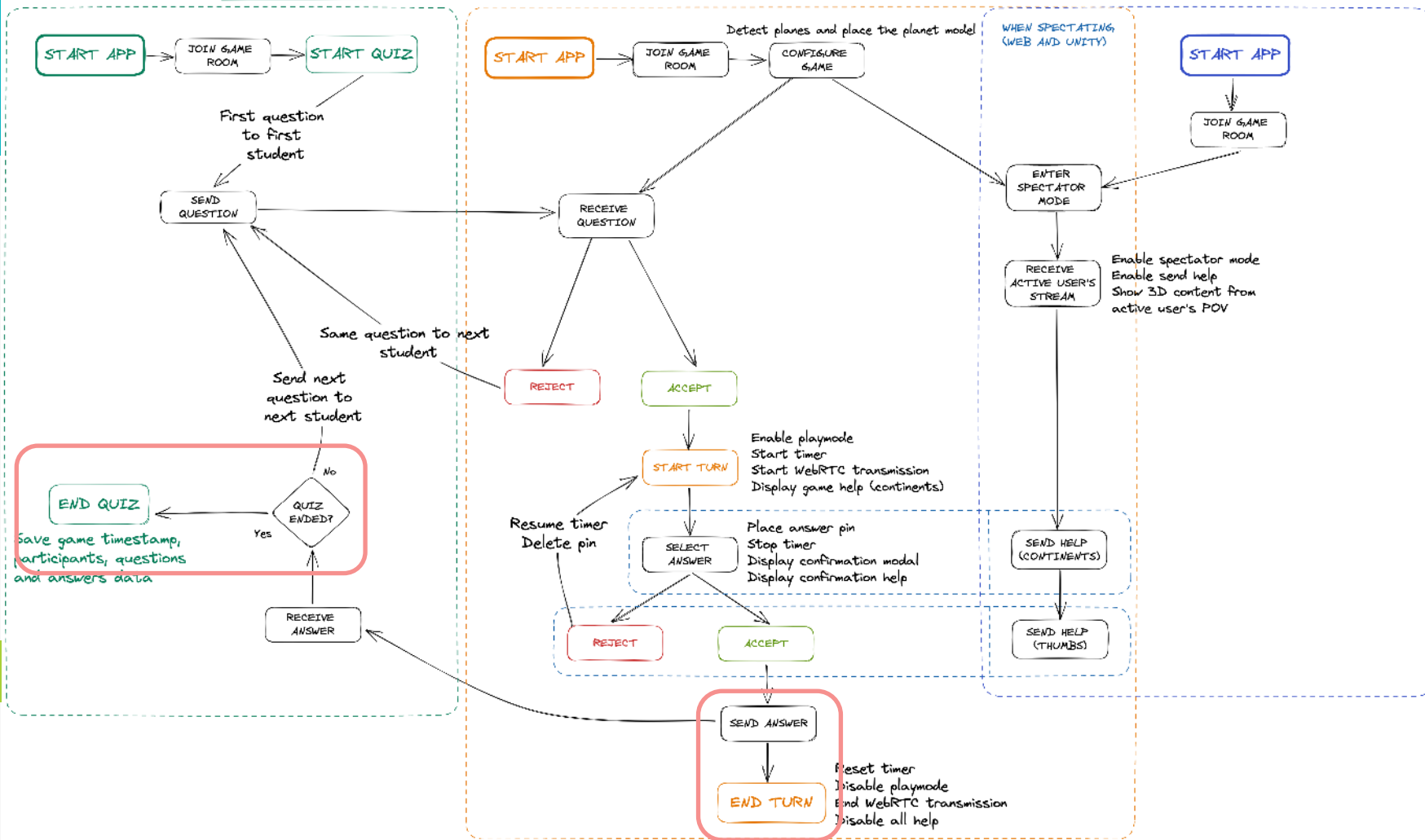
UNITY APP (STUDENT)

WEB APP (STUDENT)



WEB APP (TEACHER)UNITY APP (STUDENT)WEB APP (STUDENT)

WEB APP (TEACHER)UNITY APP (STUDENT)WEB APP (STUDENT)

WEB APP (TEACHER)UNITY APP (STUDENT)WEB APP (STUDENT)

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

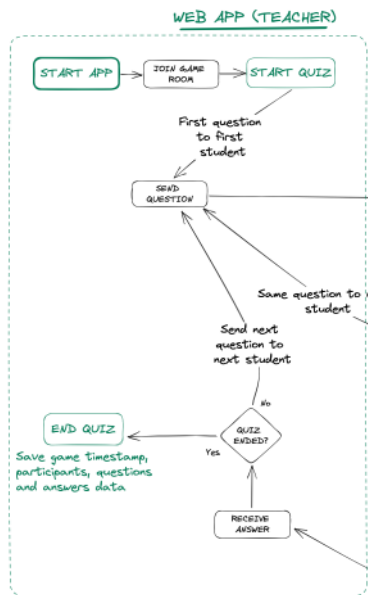
RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work



INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

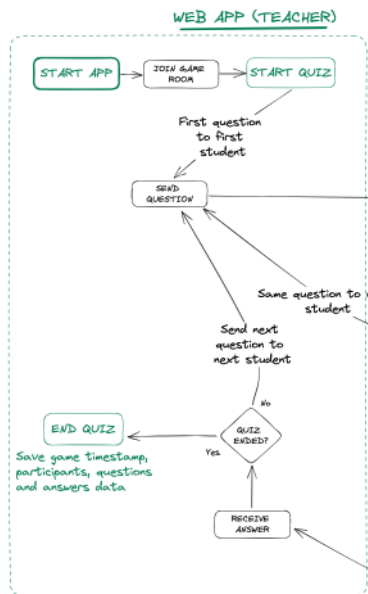
RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

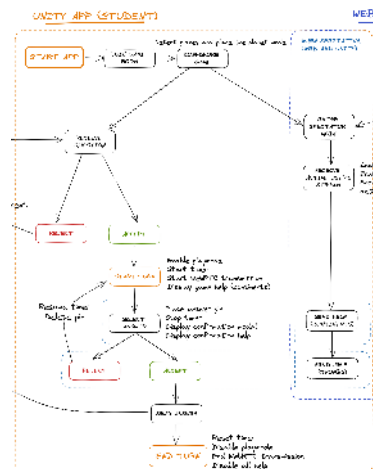
Conclusions

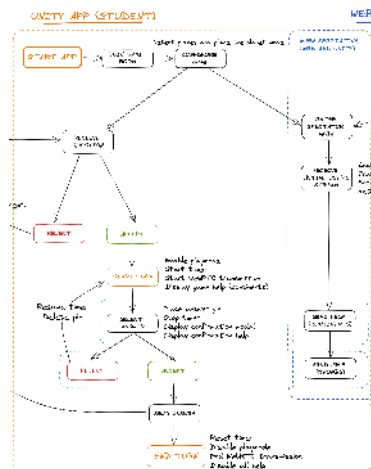
Publications

Future work



Teacher Interface





Mobile/Tablet Interface

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

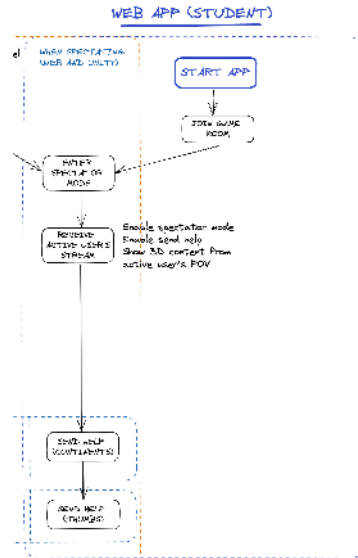
Interoperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work



INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative
AR in education

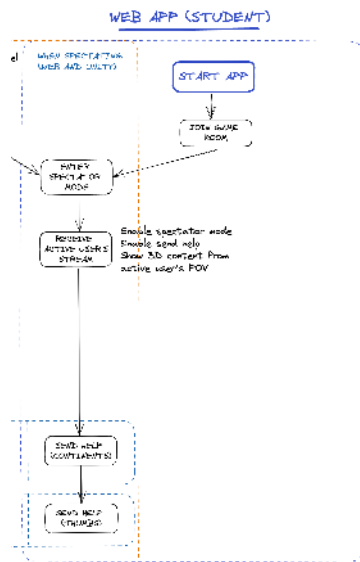
Validation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work



Web Interface

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

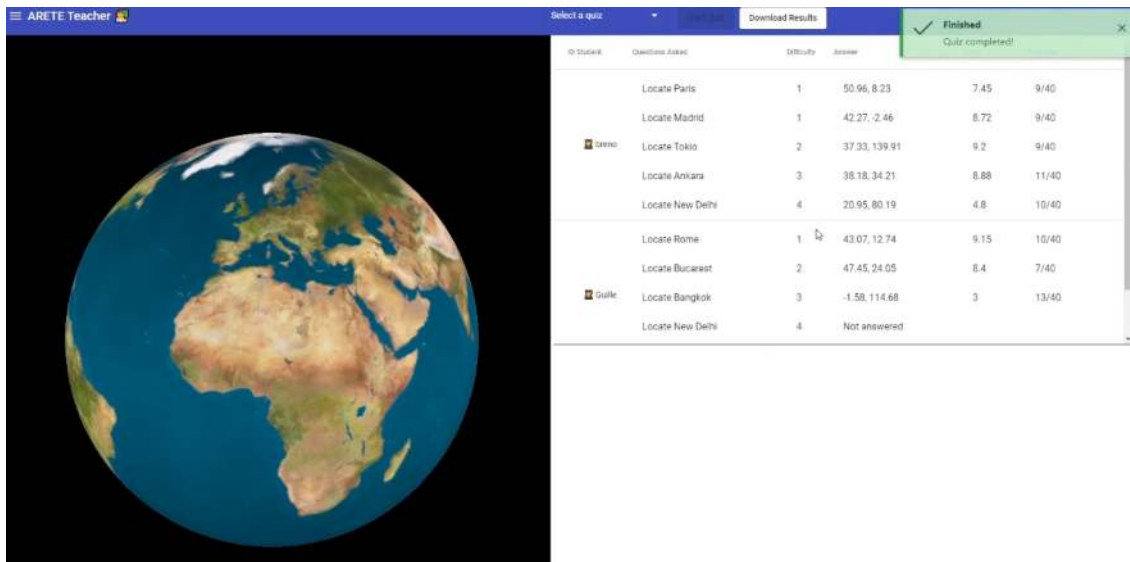
CONCLUSIONS

Conclusions

Publications

Future work

- At the end of the quiz, the teacher can **check the results**, access a **dashboard with user statistics** or **launch a new quiz**



The screenshot shows the ARETE Teacher interface. On the left is a large globe. On the right is a table displaying quiz results. The table has columns for ID, Question, Difficulty, Answer, and a completion status. A green 'Finished' banner at the top right indicates the quiz is completed.

ID	Question	Difficulty	Answer	Completion
1	Locate Paris	50.96, 8.23	7.45	9/40
1	Locate Madrid	42.27, 2.46	8.72	9/40
2	Locate Tokio	37.33, 139.91	9.2	9/40
3	Locate Ankara	38.18, 34.21	8.88	11/40
4	Locate New Delhi	20.95, 80.19	4.8	10/40
1	Locate Rome	43.07, 12.74	9.15	10/40
2	Locate Bucarest	47.45, 24.05	8.4	7/40
3	Locate Bangkok	-1.58, 114.68	3	13/40
4	Locate New Delhi	Not answered		

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

**STATE OF THE
ART**

Overview
Methodology
Results
Discussion

**RESEARCH
RESULTS**

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ **Interoperability** (D01): supports iOS and Android devices, HoloLens HMD and PC
- ▶ **Multi-user support** (D02): *Orkestra* functionalities
- ▶ **Data analytics** (D03, D04, D05): xAPI statements on a Learning Locker server
 - ▶ Basic data analysis functionalities provided by Learning Locker
 - ▶ Advanced EDA, visualisation and IA provided via custom Python package
- ▶ **Ease of development** (D06): interview with developer

- ▶ The evaluation was performed with 44 students from 3 different educational institutions: Salesianos, Zubiri Manteo y Universidad de Deusto

Participants	Gender		Players		Watchers	Total
	Males	Females	Tablets	Phones		
14-year-old	9	8	4	5	8	17
17-year-old	16	1	3	6	8	17
19-year-old	6	4	3	6	1	10
Total	31	13	10	17	17	44

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

01

Explain purpose and **experiment briefing**

02

Fill questionnaire (inspired on P-SUS)

03

Survey ID was the same as the user ID used for **in-app data collection**

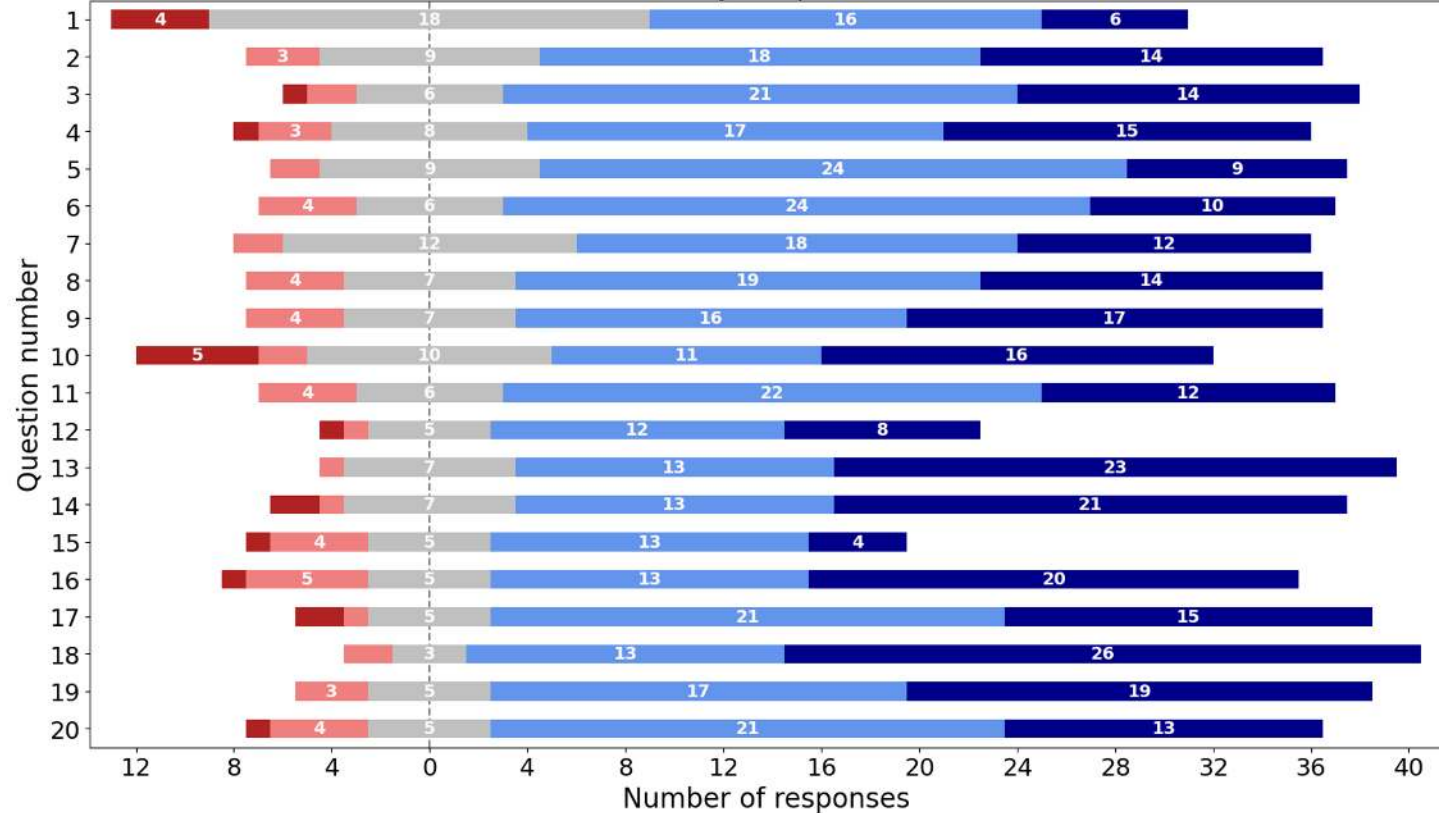
04

Post-study interview with teachers

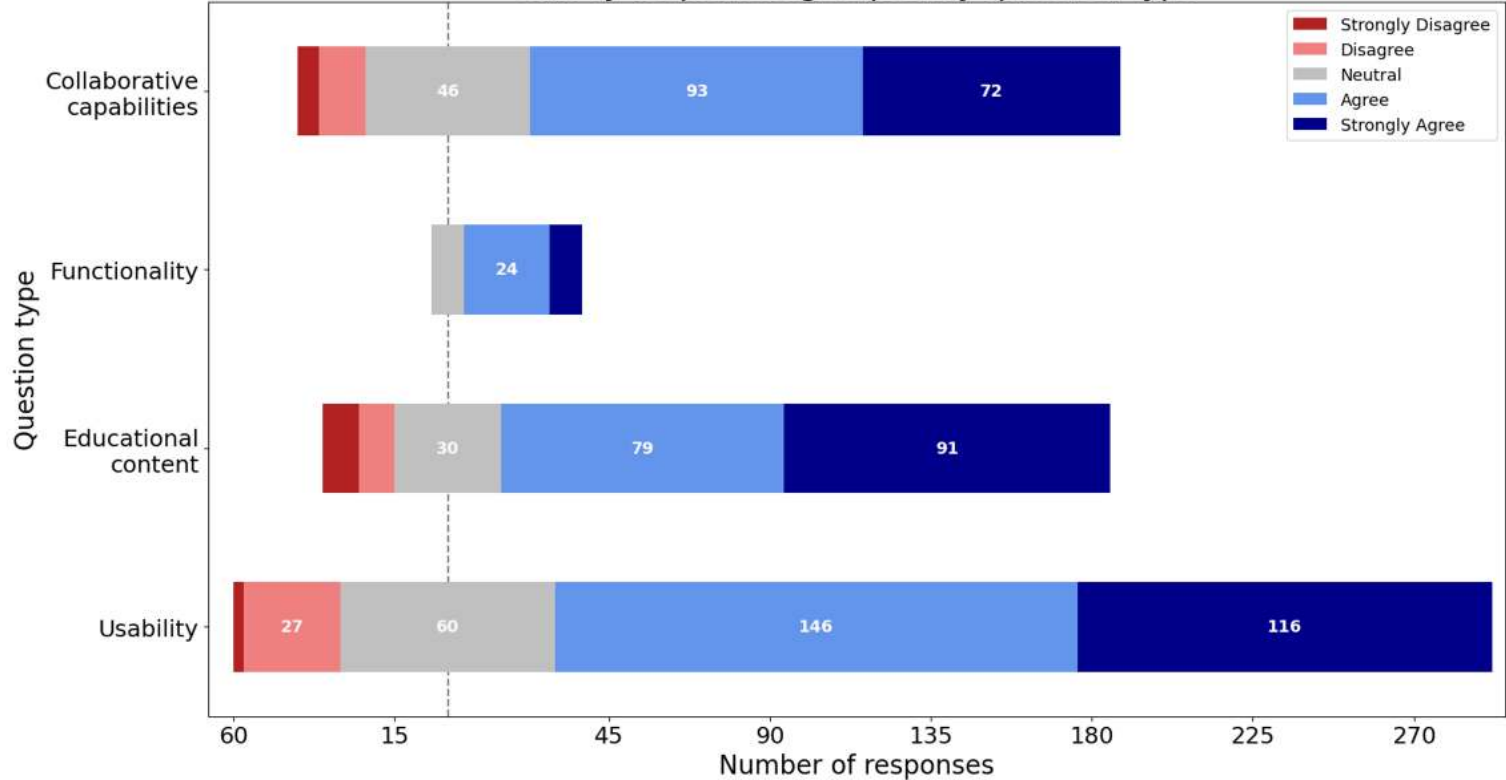
05

Data analysis from xAPI statements collected during experiment

Survey responses



Survey responses grouped by question type



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

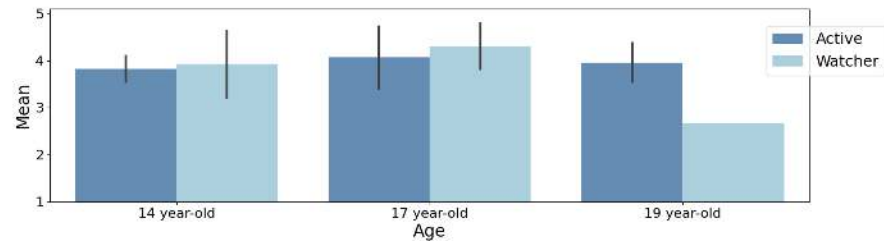
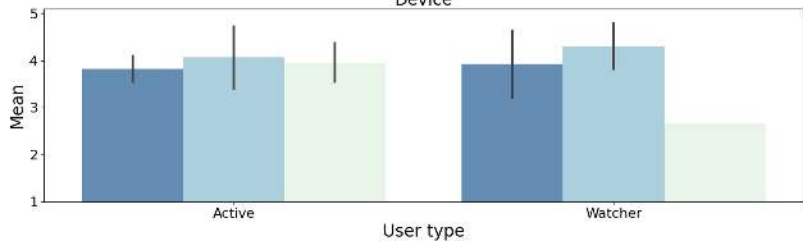
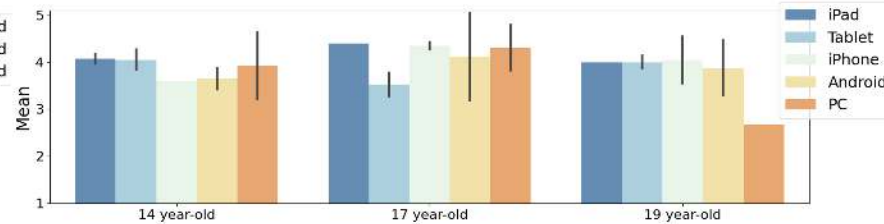
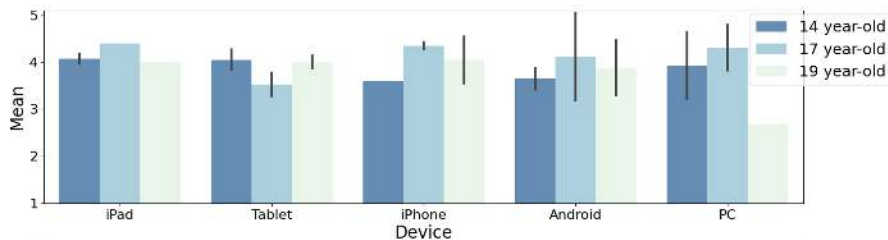
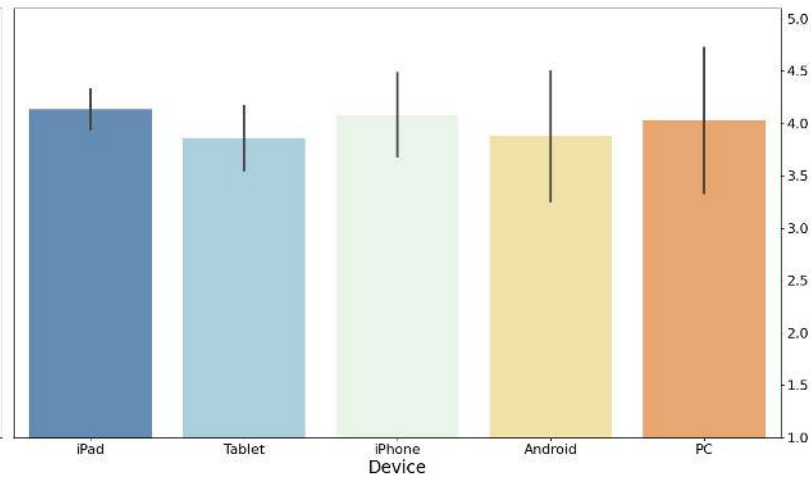
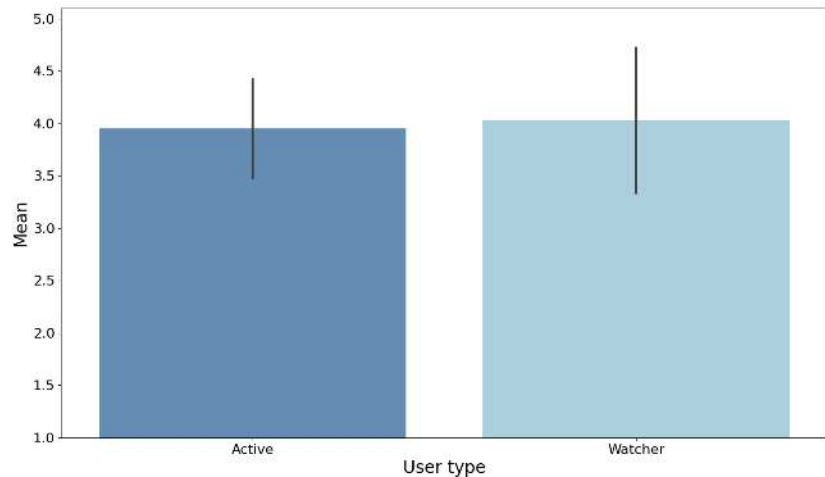
RESEARCH RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

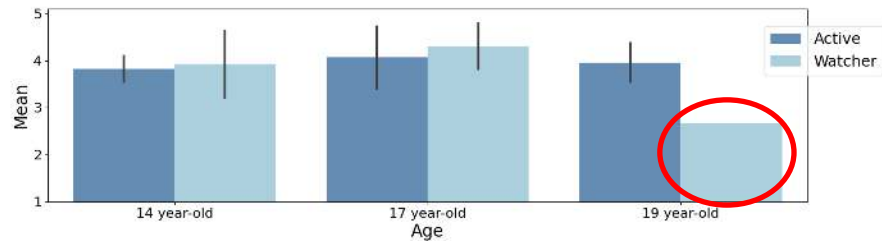
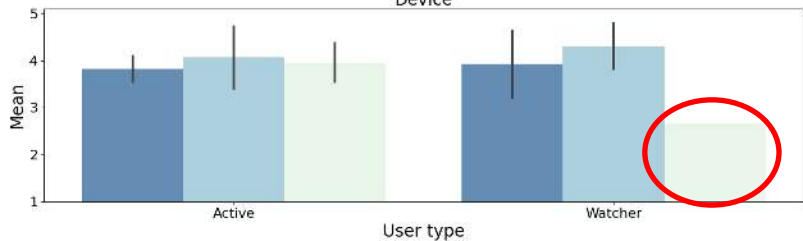
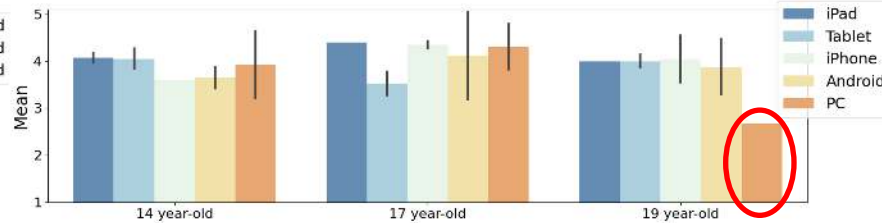
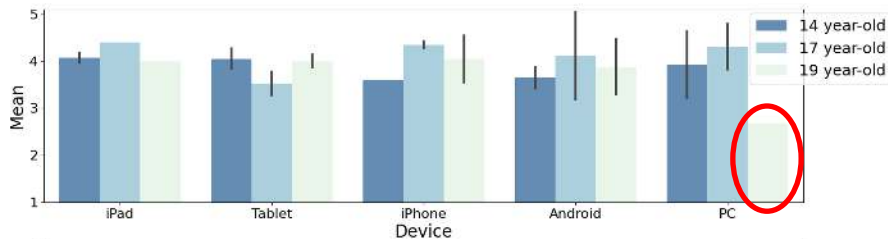
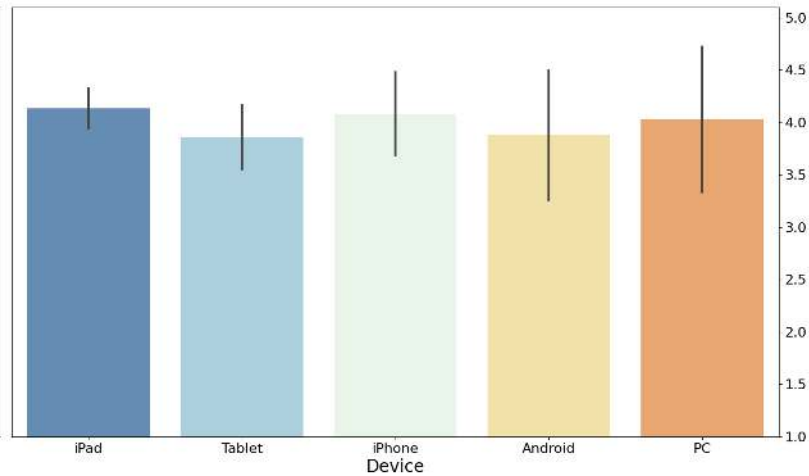
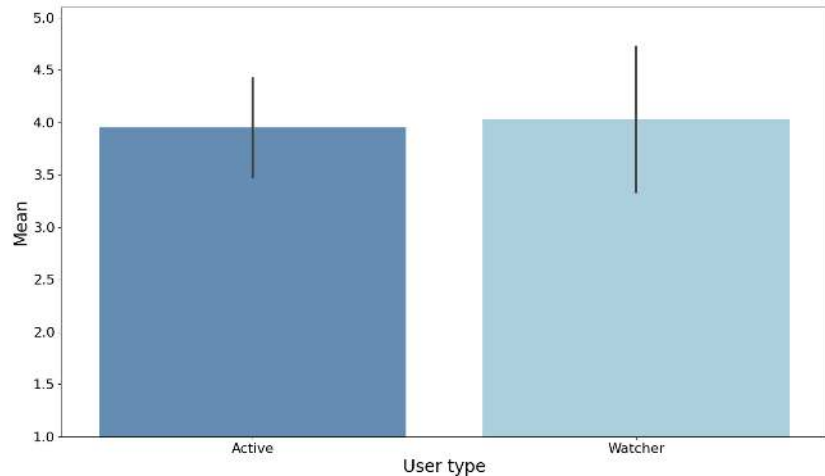
RESEARCH RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work



INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

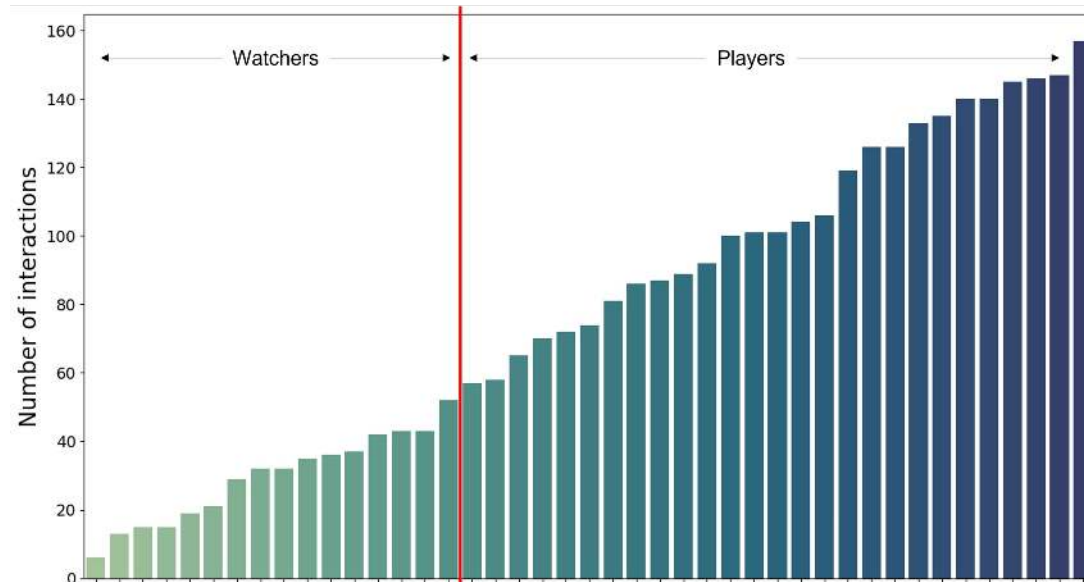
CONCLUSIONS

Conclusions

Publications

Future work

- Players interacted much more than the watchers



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Tests to detect correlations in the data and in the survey results

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Tests to detect correlations in the data and in the survey results
 - ▶ Number of interactions *vs.* survey average punctuation **X**

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Tests to detect correlations in the data and in the survey results
 - ▶ Number of interactions *vs.* survey average punctuation **X**
 - ▶ Number of interactions *vs.* quiz results **X**

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Tests to detect correlations in the data and in the survey results
 - ▶ Number of interactions *vs.* survey average punctuation ✗
 - ▶ Number of interactions *vs.* quiz results ✗
 - ▶ Survey average punctuation *vs.* number of suggestions ✓

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications
Future work

- ▶ Tests to detect correlations in the data and in the survey results
 - ▶ Number of interactions *vs.* survey average punctuation ✗
 - ▶ Number of interactions *vs.* quiz results ✗
 - ▶ Survey average punctuation *vs.* number of suggestions ✓
- ▶ Suggests that the **most engaged students** were the ones that gave a **higher score in the survey**

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

01

Collaborative features were especially appreciated

02

Personalisation options were also well received

03

Sceptical about the AI functionalities

04

Role of the *watchers* was **too passive**

05

Questions related to **exporting data** and **future updates**

4. CONCLUSIONS

Conclusions and future work

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications
Future work

01

Gaps

{ **Limited** number of works presenting **collaborative AR**
Lack of a framework providing **multi-user, multiplatform** support

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications
Future work

01

Gaps

{ **Limited** number of works presenting **collaborative AR**
Lack of a framework providing **multi-user, multiplatform** support

02

Collaboration with teachers from Ikastolen Elkartea identified **11 requirements for AR applications** to be successfully used in education

01

Gaps

{ **Limited** number of works presenting **collaborative AR**
Lack of a framework providing **multi-user, multiplatform** support

02

Collaboration with teachers from Ikastolen Elkartea identified **11 requirements for AR applications** to be successfully used in education

03

Development of **cleAR, based on 6 design objectives:**

Interoperability / multi-user support / data collection / data viz / analytics / ease of development

01

Gaps { **Limited** number of works presenting **collaborative AR**
Lack of a framework providing **multi-user, multiplatform** support

02

Collaboration with teachers from Ikastolen Elkartea identified **11 requirements for AR applications** to be successfully used in education

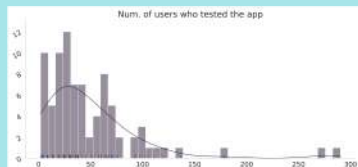
03

Development of **cleAR, based on 6 design objectives:**
Interoperability / multi-user support / data collection / data viz / analytics / ease of development

04

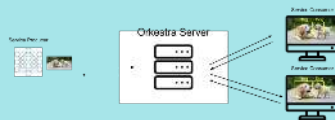
Validated by testing *ARoundTheWorld* with **44 students** from **3 educational centers**
Post-study interviews and analysis of surveys demonstrate compliance with teacher requirements

C1

In-depth analysis of the
state-of-the-art

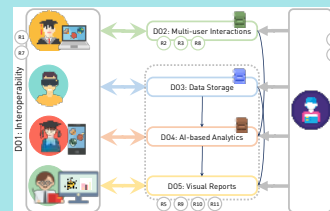
- State of research on AR applied to education
- Motivation for using AR
- Effects of AR on motivation and retention

C2

Orkestra: library for
multi-user interactions

- Low-latency
- Media synchronization
- Multiplatform

C3

cleAR: modular
architecture for AR
solutions

- Satisfies teacher requirements
- Simplifies creation of collaborative applications

C4

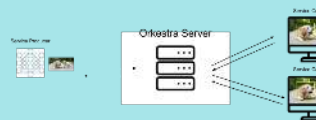
ARoundTheWorld:
implementation of an AR
app based on cleAR

- Tested in 3 educational institutions
- Validates cleAR design objectives

C1

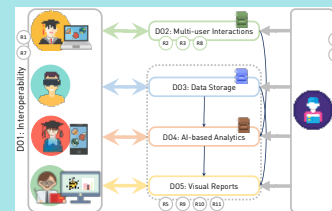
In-depth analysis of the
state-of-the-art

C2

Orkestra: library for
multi-user interactions

- Low-latency
- Media synchronization
- Multiplatform

C3

cleAR: modular
architecture for AR
solutions

- Satisfies teacher requirements
- Simplifies creation of collaborative applications

C4

ARoundTheWorld:
implementation of an AR
app based on cleAR

- Tested in 3 educational institutions
- Validates cleAR design objectives

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

CONCLUSIONS

Conclusions

Publications

Future work

Journal of Universal Computer Science, vol. 28, no. 6 (2022), 564-596
 submitted: 14-10-2021, accepted: 30-3-2022, appeared: 18-6-2022 CC BY-NC 4.0

Interactive, Collaborative and Multi-user Augmented Reality Applications in Primary and Secondary Education. A Systematic Review

Stefano Masneri

(Vicomtech, Basque Research and Technology Alliance, San Sebastian, Spain
<https://orcid.org/0000-0003-1965-5704>, s.masneri001@ik4.es)

Ana Domínguez

(Vicomtech, Basque Research and Technology Alliance, San Sebastian, Spain
<https://orcid.org/0000-0002-9329-4225>, adominguez@vicomtech.org)

Mikel Zorrilla

(Vicomtech, Basque Research and Technology Alliance, San Sebastian, Spain
<https://orcid.org/0000-0003-2589-2490>, mzorrilla@vicomtech.org)

Mikel Larrañaga

(Computer Languages and Systems, University of the Basque Country (UPV/EHU), San Sebastian, Spain
<https://orcid.org/0000-0001-9727-1197>, mikel.larrañaga@ehu.es)

Ana Arruarte

(Computer Languages and Systems, University of the Basque Country (UPV/EHU), San Sebastian, Spain
<https://orcid.org/0000-0001-5361-6640>, a.arruarte@ehu.es)

Abstract: Augmented reality is a technology that enhances human perception with additional, artificially generated sensory inputs. This creates new experiences enriching human vision by combining natural with digital elements. Augmented reality development dates back to the early sixties but it is only in the last decade, thanks to improvements to hardware and software, when it has begun to be rapidly incorporated in several fields, including education. This study presents a systematic review of the literature on the use of augmented reality applications in primary and secondary schools, with a specific focus on collaborative, multi-user and interactive applications. The aim of the study is to investigate the characteristics of such applications, the processes that led to their adoption, and their effectiveness in enhancing the learning experience. This study synthesises a set of 100 publications from 2015 to 2020 and performs a qualitative analysis of their content. The review describes the current state of the art in research in augmented reality for education and provides future research lines, as well as trends for the future of such applications in educational settings, analysing the relevance of the multi-user interaction challenge within the augmented reality ecosystem.

Keywords: Augmented Reality, Education, Literature Review, Interactive Technologies, Multi-user applications, Collaborative
Categories: H.5.1, H.5.2, K.3.1
DOI: 10.3897/jucs.76535

C3

C4

Masneri, S., Domínguez, A., Zorrilla, M., Larrañaga, M., & Arruarte, A. (2022). Interactive, collaborative and multi-user augmented reality applications in primary and secondary education. A systematic review. JUCS-Journal of Universal Computer Science, 28(6), 564-590. <https://doi.org/10.3897/jucs.76535>

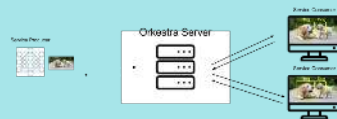
- Satisfies teacher requirements
- Simplifies creation of collaborative applications

- Tested in 3 educational institutions
- Validates cleAR design objectives

C1

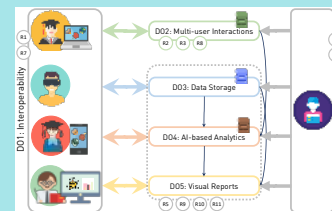
In-depth analysis of the
state-of-the-art

C2

Orkestra: library for
multi-user interactions

- Low-latency
- Media synchronization
- Multiplatform

C3

cleAR: modular
architecture for AR
solutions

- Satisfies teacher requirements
- Simplifies creation of collaborative applications

C4

ARoundTheWorld:
implementation of an AR
app based on cleAR

- Tested in 3 educational institutions
- Validates cleAR design objectives

C1

In-depth analysis of the
state-of-the-art

Pub. 1



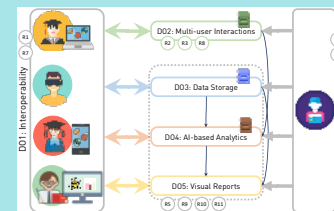
C2

Orkestra: library for
multi-user interactions

Pub. 2



C3

cleAR: modular
architecture for AR
solutions

- Satisfies teacher requirements
- Simplifies creation of collaborative applications

C4

ARoundTheWorld:
implementation of an AR
app based on cleAR

- Tested in 3 educational institutions
- Validates cleAR design objectives

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

CONCLUSIONS

Conclusions

Publications

Future work

Collaborative Multi-user Augmented Reality Solutions in the Classroom

Stefano Masneri^{1,2}, Ana Domínguez¹, Miguel Sanz¹, Iñigo Tamayo¹, Mikel Zorrilla¹, Mikel Larrañaga², and Ana Arruarte²

¹ Vicentech Foundation, Basque Research and Technology Alliance (BRTA), San Sebastián, Spain

smasneri@vicentech.org

² University of the Basque Country (UPV/EHU), San Sebastián, Spain

Abstract. Several educational institutions routinely recommend and use Augmented Reality in learning and teaching, for its potential benefits regarding an increase in the students' motivation and attention as well as their knowledge retention. Most of the time, though, augmented reality applications consist of rather static, single-user experiences which do not fuel collaborative learning. In this paper, we present a novel augmented reality platform whose aim is to facilitate collaboration and interaction between teachers and students by adding synchronization and multi-user capabilities. We present a case study demonstrating the capabilities of this platform for student-teacher and student-student collaboration.

Keywords: augmented reality, collaborative learning, multi-user applications

1 Introduction

Augmented Reality (AR) is a technology which overlays virtual objects (also called 'augmentations', see [1]) onto the real world, and the virtual objects co-exist in the same space as objects in the real world [2]. Despite AR first being introduced in the early nineties as a performance aid for manufacturing airplanes [3], and with precursors dating back several more decades, it is only in recent years that AR applications are more broadly adopted by both industry and academia. This is due to the availability of libraries supporting AR development and the decreasing costs of the required hardware. Especially in education, high-end equipment such as head-mounted displays (HMD) are still to be missed largely, while tablets and smartphones allow to routinely use AR technology in the classroom, accessible by a broader audience.

In the context of Technology Enhanced Learning (TEL), AR provides a multi-modal learning environment, capable of improving users' experiences and increasing the students' motivation [4]. The most relevant advantages of successfully using AR-based applications in the classroom in comparison with traditional learning are [5]:

- Decrease of students' cognitive load

Masneri, S., Domínguez, A., Sanz, M., Tamayo, I., Zorrilla, M., Larrañaga, M., & Arruarte, A. (2022). Collaborative Multi-user Augmented Reality Solutions in the Classroom. In: Auer, M.E., Hortsch, H., Michler, O., Köhler, T. (eds) Mobility for Smart Cities and Regional Development - Challenges for Higher Education. ICL 2021. Lecture Notes in Networks and Systems, vol 390. Springer, Cham. https://doi.org/10.1007/978-3-030-93907-6_106

- Simplifies creation of collaborative applications

- Validates cleAR design objectives

C1

In-depth analysis of the
state-of-the-art

Pub. 1

Hochreiter, C. Collaborative and Artificially-Augmented Reality Applications in Primary and Secondary Education: A Systematic Review

Abstract: This paper presents a systematic review of the state-of-the-art in the field of collaborative and artificially-augmented reality (AR) applications in primary and secondary education. The review covers the period from 2010 to 2020 and includes 100 papers. The results show that the majority of the research is focused on the development of AR applications for educational purposes, with a focus on the use of AR in the classroom. The review also identifies the challenges and opportunities in the field of collaborative and artificially-augmented reality applications in primary and secondary education.

Keywords: Collaborative, Artificially-Augmented Reality, Primary and Secondary Education, Systematic Review

1. Introduction

2. Methodology

3. Results

4. Discussion

5. Conclusion

References

1. Introduction

2. Methodology

3. Results

4. Discussion

5. Conclusion

References

C2

Orkestra: library for
multi-user interactions

Pub. 2

C2: alternative to multi-user Augmented Reality Solutions in the Classroom

Abstract: This paper presents a library for multi-user interactions in the classroom. The library is designed to be used in conjunction with the Orkestra framework, which is a framework for multi-user interactions in the classroom. The library provides a set of APIs that can be used to create multi-user interactions in the classroom. The library is designed to be easy to use and to integrate with existing classroom applications.

Keywords: Multi-user Interactions, Augmented Reality, Classroom, Library

1. Introduction

2. Methodology

3. Results

4. Discussion

5. Conclusion

References

1. Introduction

2. Methodology

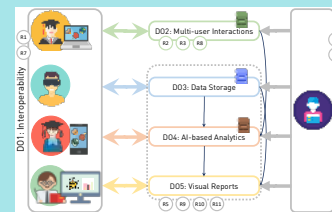
3. Results

4. Discussion

5. Conclusion

References

C3

cleAR: modular
architecture for AR
solutions

- Satisfies teacher requirements
- Simplifies creation of collaborative applications

C4

ARoundTheWorld:
implementation of an AR
app based on cleAR

- Tested in 3 educational institutions
- Validates cleAR design objectives

C1

In-depth analysis of the
state-of-the-art

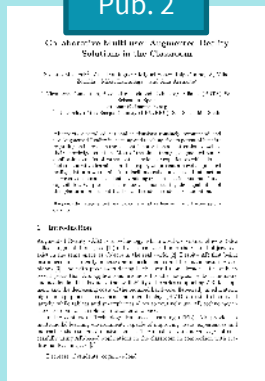
Pub. 1



C2

Orkestra: library for
multi-user interactions

Pub. 2



C3

cleAR: modular
architecture for AR
solutions

Pub. 3



C4

ARoundTheWorld:
implementation of an AR
app based on cleAR

- Tested in 3 educational institutions
- Validates cleAR design objectives

**cleAR: an interoperable architecture for multi-user AR-based school curricula**Stefano Masneri¹ · Ana Domínguez² · Miguel Sanz² · Mikel Zorrilla² · Mikel Larrañaga¹ · Ana Arruarte¹Received: 4 February 2022 / Accepted: 28 January 2023 / Published online: 27 February 2023
© The Author(s) 2023, corrected publication 2023**Abstract**

Although there are some experiences that demonstrate the validity of the use of augmented reality in schools to help students understand and retain complex concepts, augmented reality has not been widely adopted in the education sector yet. This is in part because it is hard to use augmented reality applications in collaborative learning scenarios and to integrate them in the existing school curricula. In this work, we present an interoperable architecture that simplifies the creation of augmented reality applications, enables multi-user student collaboration and provides advanced mechanisms for data analysis and visualization. A review of the literature together with a survey answered by 47 primary and secondary school teachers allowed us to identify the design objectives of cleAR, an architecture for augmented reality-based collaborative educational applications. cleAR has been validated through the development of three proofs of concept. cleAR provides a more mature technological ecosystem that will foster the emergence of augmented reality applications for education and their inclusion in existing school programs.

Keywords Augmented reality · Multi-user interactions · Collaborative learning · Visual learning analytics · Artificial intelligence

1 Introduction

The usage of technology in the last decade has transformed the education sector, empowering teachers and students by creating more engaging environments, facilitating collaboration and enabling different learning styles. During the

COVID-19 pandemic and its associated lockdowns, most teaching activities have moved online and all stakeholders relied on different technologies to keep school activities ongoing (Azorin 2020).

In recent years, augmented reality (AR) technology is being used more and more, thanks to an ever-increasing number of devices supporting it, as well as a more mature software ecosystem that allows developers to speedup the creation of AR-based applications. With their ability to attach virtual content to any physical surface, either through the usage of markers or by using plane detection or geographical information, AR applications have found a valuable role in training and education. Several companies offer educational AR apps and many scientific publications have shown that AR can enhance and improve the learning experience (Akçayır and Akçayır 2017).

Unfortunately though, AR has not yet seen widespread usage in education, and most of the experiences carried out so far are based on small experiments. This is due to several causes (presented in Sect. 3.2) which can be summarized with the difficulty for developers and educators to create content that can be used by every student and that integrates well with the existing school curricula. One of the

✉ Stefano Masneri
smasneri001@ikasle.ehu.es
Ana Domínguez
adominguez@vicomtech.org
Miguel Sanz
msanz@vicomtech.org
Mikel Zorrilla
mzorilla@vicomtech.org
Mikel Larrañaga
mikel.larrañaga@ehu.es
Ana Arruarte
aaruarte@ehu.es

¹ Computer Languages and Systems Department, University of the Basque Country UPV/EHU, San Sebastián, Spain

² Fundación Vicomtech, Basque Research and Technology Alliance (BRTA), San Sebastián, Spain

C3

cleAR: modular

C4

ARoundTheWorld:

Masneri, S., Domínguez, A., Sanz, M., Zorrilla, M., Larrañaga, M., & Arruarte, A. (2023). **cleAR: an interoperable architecture for multi-user AR-based school curricula**. *Virtual Reality* 27, 1813–1825 (2023). <https://doi.org/10.1007/s10055-023-00764-5>

- Tested in 3 educational institutions
- Validates cleAR design objectives

C1

In-depth analysis of the state-of-the-art

Pub. 1

Hochschule, Collaboration zur Entwicklung von AR-angebots
 Real-time-Rendering in Primary and Secondary
 Education

Abstract: This paper presents a comprehensive analysis of the state-of-the-art in AR applications for primary and secondary education. The analysis is based on a systematic literature review of 100 peer-reviewed articles published between 2010 and 2020. The results show that AR applications are increasingly being used in primary and secondary education, with a focus on interactive learning and collaboration. The analysis also identifies key challenges and opportunities for the future of AR in education.

Keywords: AR, Education, Primary, Secondary, Interactive Learning, Collaboration

1. Introduction
 AR applications have gained significant attention in recent years, particularly in the field of education. This paper provides a comprehensive analysis of the state-of-the-art in AR applications for primary and secondary education.

2. Literature Review
 A systematic literature review was conducted to identify and analyze the most relevant research in this field. The review included 100 peer-reviewed articles published between 2010 and 2020.

3. Results
 The results of the literature review show that AR applications are increasingly being used in primary and secondary education. The most common applications are interactive learning and collaboration.

4. Challenges and Opportunities
 The analysis also identifies key challenges and opportunities for the future of AR in education. The challenges include the need for more research and development, and the need for more training and support for teachers.

5. Conclusion
 The analysis shows that AR applications are increasingly being used in primary and secondary education. The results also identify key challenges and opportunities for the future of AR in education.

6. References
 [1] Smith, J. (2010). AR applications in education. *Journal of Educational Technology*, 12(1), 1-10.

[2] Jones, K. (2011). AR applications in education. *Journal of Educational Technology*, 13(2), 1-10.

[3] Brown, L. (2012). AR applications in education. *Journal of Educational Technology*, 14(3), 1-10.

[4] Green, M. (2013). AR applications in education. *Journal of Educational Technology*, 15(4), 1-10.

[5] White, N. (2014). AR applications in education. *Journal of Educational Technology*, 16(5), 1-10.

[6] Black, O. (2015). AR applications in education. *Journal of Educational Technology*, 17(6), 1-10.

[7] Grey, P. (2016). AR applications in education. *Journal of Educational Technology*, 18(7), 1-10.

[8] Brown, L. (2017). AR applications in education. *Journal of Educational Technology*, 19(8), 1-10.

[9] Green, M. (2018). AR applications in education. *Journal of Educational Technology*, 20(9), 1-10.

[10] White, N. (2019). AR applications in education. *Journal of Educational Technology*, 21(10), 1-10.

C2

Orkestra: library for multi-user interactions

Pub. 2

Orkestra: A Library for Multi-User Interactions in AR

Abstract: This paper presents Orkestra, a library for multi-user interactions in AR. Orkestra is designed to facilitate the development of AR applications that support collaborative learning and interaction. The library is based on a set of principles that guide the design of AR applications for multi-user interactions.

Keywords: AR, Multi-User Interactions, Collaborative Learning, Interaction Design

1. Introduction
 AR applications have gained significant attention in recent years, particularly in the field of education. This paper presents Orkestra, a library for multi-user interactions in AR.

2. Design Principles
 Orkestra is designed to facilitate the development of AR applications that support collaborative learning and interaction. The library is based on a set of principles that guide the design of AR applications for multi-user interactions.

3. Architecture
 Orkestra is designed to facilitate the development of AR applications that support collaborative learning and interaction. The library is based on a set of principles that guide the design of AR applications for multi-user interactions.

4. Implementation
 Orkestra is designed to facilitate the development of AR applications that support collaborative learning and interaction. The library is based on a set of principles that guide the design of AR applications for multi-user interactions.

5. Evaluation
 Orkestra is designed to facilitate the development of AR applications that support collaborative learning and interaction. The library is based on a set of principles that guide the design of AR applications for multi-user interactions.

6. Conclusion
 Orkestra is designed to facilitate the development of AR applications that support collaborative learning and interaction. The library is based on a set of principles that guide the design of AR applications for multi-user interactions.

7. References
 [1] Smith, J. (2010). AR applications in education. *Journal of Educational Technology*, 12(1), 1-10.

[2] Jones, K. (2011). AR applications in education. *Journal of Educational Technology*, 13(2), 1-10.

[3] Brown, L. (2012). AR applications in education. *Journal of Educational Technology*, 14(3), 1-10.

[4] Green, M. (2013). AR applications in education. *Journal of Educational Technology*, 15(4), 1-10.

[5] White, N. (2014). AR applications in education. *Journal of Educational Technology*, 16(5), 1-10.

[6] Black, O. (2015). AR applications in education. *Journal of Educational Technology*, 17(6), 1-10.

[7] Grey, P. (2016). AR applications in education. *Journal of Educational Technology*, 18(7), 1-10.

[8] Brown, L. (2017). AR applications in education. *Journal of Educational Technology*, 19(8), 1-10.

[9] Green, M. (2018). AR applications in education. *Journal of Educational Technology*, 20(9), 1-10.

[10] White, N. (2019). AR applications in education. *Journal of Educational Technology*, 21(10), 1-10.

C3

cleAR: modular architecture for AR solutions

Pub. 3

cleAR: A Modular Architecture for AR Solutions

Abstract: This paper presents cleAR, a modular architecture for AR solutions. cleAR is designed to facilitate the development of AR applications that support collaborative learning and interaction. The architecture is based on a set of principles that guide the design of AR applications for multi-user interactions.

Keywords: AR, Modular Architecture, Collaborative Learning, Interaction Design

1. Introduction
 AR applications have gained significant attention in recent years, particularly in the field of education. This paper presents cleAR, a modular architecture for AR solutions.

2. Design Principles
 cleAR is designed to facilitate the development of AR applications that support collaborative learning and interaction. The architecture is based on a set of principles that guide the design of AR applications for multi-user interactions.

3. Architecture
 cleAR is designed to facilitate the development of AR applications that support collaborative learning and interaction. The architecture is based on a set of principles that guide the design of AR applications for multi-user interactions.

4. Implementation
 cleAR is designed to facilitate the development of AR applications that support collaborative learning and interaction. The architecture is based on a set of principles that guide the design of AR applications for multi-user interactions.

5. Evaluation
 cleAR is designed to facilitate the development of AR applications that support collaborative learning and interaction. The architecture is based on a set of principles that guide the design of AR applications for multi-user interactions.

6. Conclusion
 cleAR is designed to facilitate the development of AR applications that support collaborative learning and interaction. The architecture is based on a set of principles that guide the design of AR applications for multi-user interactions.

7. References
 [1] Smith, J. (2010). AR applications in education. *Journal of Educational Technology*, 12(1), 1-10.

[2] Jones, K. (2011). AR applications in education. *Journal of Educational Technology*, 13(2), 1-10.

[3] Brown, L. (2012). AR applications in education. *Journal of Educational Technology*, 14(3), 1-10.

[4] Green, M. (2013). AR applications in education. *Journal of Educational Technology*, 15(4), 1-10.

[5] White, N. (2014). AR applications in education. *Journal of Educational Technology*, 16(5), 1-10.

[6] Black, O. (2015). AR applications in education. *Journal of Educational Technology*, 17(6), 1-10.

[7] Grey, P. (2016). AR applications in education. *Journal of Educational Technology*, 18(7), 1-10.

[8] Brown, L. (2017). AR applications in education. *Journal of Educational Technology*, 19(8), 1-10.

[9] Green, M. (2018). AR applications in education. *Journal of Educational Technology*, 20(9), 1-10.

[10] White, N. (2019). AR applications in education. *Journal of Educational Technology*, 21(10), 1-10.

C4

ARoundTheWorld: implementation of an AR app based on cleAR



- Tested in 3 educational institutions
- Validates cleAR design objectives

In-depth analysis of the state-of-the-art

DOI: 10.1002/eqm2

Orkestra: library for multi-user interactions

[illegible]

cleAR: modular
architecture for AR
solutions

[illegible]

ARoundTheWorld:
implementation of an AR
app based on cleAR



A collaborative AR application for education: from architecture design to user evaluation

Stefano Masneri¹ · Ana Dominguez² · Guillermo Pacho² · Mikel Zorrilla² · Mikel Larrañaga¹ · Ana Arruarte¹Received: 20 June 2023 / Accepted: 3 January 2024
© The Author(s) 2024

Abstract

Augmented reality applications can be used in an educational context to facilitate learning. In particular, augmented reality has been successfully used as a tool to boost students' engagement and to improve their understanding of complex topics. Despite this, augmented reality usage is still not common in schools and it still offers mostly individual experiences, lacking collaboration capabilities which are of paramount importance in a learning environment. This work presents an application called *ARoundTheWorld*, a multipatform augmented reality application for education. It is based on a software architecture, designed with the help of secondary school teachers, that provides interoperability, multi-user support, integration with learning management systems and data analytics capabilities, thus simplifying the development of collaborative augmented reality learning experiences. The application has been tested by 44 students and 3 teachers from 3 different educational institutions to evaluate the usability as well as the impact of collaboration functionalities in the students' engagement. Qualitative and quantitative results show that the application fulfils all the design objectives identified by teachers as key elements for augmented reality educational applications. Furthermore, the application was positively evaluated by the students and it succeeded in promoting collaborative behaviour. These results show that *ARoundTheWorld*, and other applications built using the same architecture, could be easily developed and successfully integrated into existing schools curricula.

Keywords Augmented reality · Multi-user interactions · Collaborative learning · User evaluation · Data analysis

1 Introduction

Augmented reality (AR) is a technology which overlays virtual objects (augmented components) into the real world. These virtual objects then appear to coexist in the same space as objects in the real world (Akçayir and Akçayir 2017; Azuma et al. 2001). Another widely used definition is from Milgram and Kishino (1994) where, in a continuum ranging from a purely virtual environment to a completely real one, AR is positioned close to the real environment and the users perceive the real world with an additional layer of virtuality. Even though the technology was originally used as a tool to help assembly workers at Boeing by showing them virtual labels through the use of custom made glasses (Caulfield and Mizell 1992), AR has quickly attracted research and industry attention in many different areas such as gaming (Das et al. 2017), cultural heritage (Vlachos et al. 2022), customer engagement (McLean and Wilson 2019), manufacturing (Ong et al. 2008) or education (Aytaç-Gürzon et al. 2021; Garzón et al. 2019). Especially, in the last ten years, there has been a surge of research related to the usage

✉ Stefano Masneri
smasneri01@ikasle.ehu.es
Ana Dominguez
adominguez@vicomtech.org
Guillermo Pacho
gacho@vicomtech.org
Mikel Zorrilla
mzorrilla@vicomtech.org
Mikel Larrañaga
mikel.larrañaga@ehu.es
Ana Arruarte
aarruarte@ehu.es

¹ Computer Languages and Systems Department, University of the Basque Country UPV/EHU, San Sebastián, Spain

² Fundación Vicomtech, Basque Research and Technology Alliance (BRTA), San Sebastián, Spain

C3

cleAR: modular

C4

ARoundTheWorld:

Masneri, S., Domínguez, A., Pacho, G., Zorrilla, M., Larrañaga, M., Arruarte, A. A collaborative AR application for education: from architecture design to user evaluation. *Virtual Reality* 28, 42 (2024). <https://doi.org/10.1007/s10055-024-00952-x>

In-depth analysis of the state-of-the-art

**Innovative Collaborative e-Learning Integrated
Health Applications in Primary and Secondary
Education:
A Systematic Review**

[illegible]

Abstract: The effects of the combination of the two most common human immunodeficiency virus (HIV) transmission routes, heterosexual intercourse and injection drug use, on the risk of HIV infection were examined. Data from a cohort study of 1,000 injection drug users in San Francisco, California, who were followed for 10 years, were analyzed. The risk of HIV infection was estimated using a multivariate model that included variables for sexual and injection risk factors. The results showed that the combination of sexual and injection risk factors significantly increased the risk of HIV infection. The risk of HIV infection was highest for individuals who had both sexual and injection risk factors. The results suggest that the combination of these two risk factors is a major factor in the transmission of HIV.

Orkestra: library for multi-user interactions

City-shares built new Anguierre. The by
Solutions in the Classroom.

© 2000 Blackwell Science Ltd, *Journal of Internal Medicine* 247: 369–376

2. Introduction

[illegible]

cleAR: modular
architecture for AR
solutions

doi:10.1016/j.sbspro.2012.09.005

Enrique Alvarado^{1,2}, Juan Camargo², Miguel Lora², Mikel Zavala², Mikel Camargo², Ana Briceño²

[illegible][illegible]

ARoundTheWorld: implementation of an AR app based on cleAR

ORIGINAL ARTICLE

A collaborative AR application for education: from architecture design

Enrique Almaraz¹, Ana Dominguez², Guillermo Pineda³, Wilfrido Zavala⁴, Wilfrido Lavatanga⁵, Ana Domínguez

Abstract The present study was designed to assess the relationship between organizational commitment and organizational citizenship behavior (OCB) in the workplace. The study was conducted in a large, multi-national organization. The results of the study indicate that organizational commitment is a significant predictor of OCB. The study also found that organizational commitment is a significant predictor of OCB in the workplace. The study also found that organizational commitment is a significant predictor of OCB in the workplace. The study also found that organizational commitment is a significant predictor of OCB in the workplace.

the same authors.

1 Introduction

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

Publications related to ARETE project

Publications related to ARETE project

Work-in-progress—ARETE - An Interactive Educational System using Augmented Reality

Radford Morgan Elementary Education University of the State College 100 University Ave. New Auburn, MN 56007 morgandr@uwc.edu	Jon D. Boudreau 2nd Grade Coordinator George Brown's 2nd and Technology Building St. Anthony's School 2000 University Ave. N St. Anthony, MN 55005	Johnnie Wal Director of Educational Services 1000 University Ave. St. Anthony, MN 55005 johnnie.wal@uwc.edu
Janice Popp Health, Behavioral and Wellness Services Four University University of the State College 100 University Ave. St. Anthony, MN 55005	Stephanie Hines Office of Technology University of the State College 100 University Ave. St. Anthony, MN 55005	Janette Tiedt Chief of Student Programs University of the State College 100 University Ave. St. Anthony, MN 55005
Julia Wray Instructional Technology St. Anthony's 2000 University Ave. St. Anthony, MN 55005	Gregory Ullmann Office of the President University of the State College 100 University Ave. St. Anthony, MN 55005	David Mangione Head of Campus Safety University of the State College 100 University Ave. St. Anthony, MN 55005

[illegible]

3. Discussion

Integrated BMD software offers to the customer digital copies of information from photocranial radiographs. It is important to note that the BMD software does not generate a digital copy of the original radiograph. The original radiograph must be available to the user to allow for the generation of the digital copy. The original radiograph must be available to the user to allow for the generation of the digital copy. The original radiograph must be available to the user to allow for the generation of the digital copy.

© 2005 Blackwell Publishing Ltd, *Journal of Internal Medicine* 257: 255–262

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE ART

Overview

Methodology

Results

Discussion

RESEARCH RESULTS

Interoperable architecture for collaborative AR in education

Validation of the proposed architecture

CONCLUSIONS

Conclusions

Publications

Future work

Work-in-progress—ARETE - An Interactive Educational System using Augmented Reality

Siefano Masneri
Vicomech Foundation
University of the Basque Country
San Sebastian, Spain
smasneri@vicomech.org

Ana Domínguez
Vicomech Foundation
Basque Research and Technology Alliance
San Sebastian, Spain
adominguez@vicomech.org

Fridolin Wild
Institute of Educational Technology
The Open University
Milton Keynes, UK
f.wild@open.ac.uk

Jeroen Pronk
Faculty of Behavioral and Movement Sciences
Vrije Universiteit
Amsterdam, The Netherlands
j.pronk@vu.nl

Matthias Heintz
School of Informatics
University of Leicester
Leicester, UK
mnh21@leicester.ac.uk

Jennifer Tiede
Chair of School Pedagogy
University of Würzburg
Würzburg, Germany
jennifer.tiede@uni-wuerzburg.de

Adina Nistor
European Schoolcraft
Brussels, Belgium
adina.nistor@eun.org

Giuseppe Chiazese
Istituto per le Tecnologie Didattiche
Consiglio Nazionale delle Ricerche
Palermo, Italy
giuseppe.chiazese@itd.cnr.it

Eleni Mangina
School of Computer Science
University College Dublin (UCD)
Dublin, Ireland
eleni.mangina@ucd.ie

Abstract—ARETE (Augmented Reality Interactive Educational System) is a European project, which aims to develop an interactive toolkit for Augmented Reality (AR) content. In this work in progress paper, we describe the system which is currently being developed. The ARETE system follows human-centered interaction design practices and has a strong focus in interactive, multi-user and multi-lingual technologies. On completion, the system will be evaluated across three pilot studies. The three pilots will involve more than 3000 students across ten European countries and they will assess the impact of usage of the ARETE system and the educational value of AR for English literacy skills. STEM (Science, Technology, Engineering and Maths) skills and in the implementation of Positive Behaviour Support in Schools (PBIS). ARETE will also be involved in the implementation of new standards for the creation of cross-platform content and services as well as the creation of learning experience data repositories.

Index Terms—augmented reality, digital education, interactive systems, standardization

I. INTRODUCTION

Augmented Reality (AR) refers to the real-time digital overlay of information over physical elements [1]. Research on AR related technology as well as its industrial applications are flourishing all over the world and especially across Europe, in various domains. These application areas range from healthcare to biomedical, from agriculture to transportation, from training and education to gaming and entertainment purposes. AR has been described as a single-user tool with provision for collaborative spaces [2]. The ARETE project [3] aims to

build a collaborative AR ecosystem with multi-user interaction to add value to education.

With the ongoing spread of Covid19 pandemic [4] affecting working and learning models around the globe, the importance and need of digital education becomes more relevant than it was ever before. By targeting high technology readiness level, ARETE will strengthen the European canvas of research, industrial capacities and market acceptance for positioning it as a leader across the globe.

The ARETE project aims to develop, integrate and disseminate interactive technologies via AR methods and tools. The creation of new technologies and the inter-connection of existing digital systems will build a pan-European sustainable competitive ecosystem. Thus, the ARETE ecosystem will support fast dissemination of augmented learning content to a wide audience, while strengthening the research and industrial capacities in Europe to develop future interactive devices for education, taking into account accessibility to a broader community.

In this paper we present the work under progress since the start of the ARETE project in November 2019 as well its impact that is envisioned for the betterment of society, economy and communities from research and industry.

II. PROJECT OBJECTIVES

The overall scope of the ARETE project is defined around four main objectives as listed below:

Masneri, S., Domínguez, A., Wild, F., Pronk, J., Heintz, M., Tiede, J., Nistor, A., Chiazese, G., & Mangina, E. (2020). Work-in-progress-ARETE- An Interactive Educational System using Augmented Reality. In D. Economou, A. Klippel, H. Dodds, A. Pena-Rios, M. J. W. Lee, D. Beck, J. Pirker, A. Dengel, T. M. Peres, & J. Richter (Eds.), 2020 6th International Conference of the Immersive Learning Research Network (iLRN): Conference Proceedings (pp. 283-286). <http://doi.org/10.23919/iLRN47897.2020.9155186>

Publications related to ARETE project

Work-in-progress—ARETE - An Interactive Educational System using Augmented Reality

[illegible]

1. Introduction.

Synthetic Biology (SB) refers to the systematic design and construction of new physical components, or systems, and the subsequent use of these components to develop new biological systems. It is a multidisciplinary field that combines principles from biology, engineering, computer science, and chemistry to create artificial biological systems. These systems are designed to perform specific functions, such as producing a particular molecule, detecting a specific environmental condition, or performing a specific task. The goal of SB is to create biological systems that are more efficient, more reliable, and more predictable than natural biological systems. This field has the potential to revolutionize many areas of science and technology, including medicine, agriculture, and environmental science. In this paper, we will discuss the current state of SB, its applications, and its future prospects. We will also discuss the challenges and opportunities associated with this field.

2. The current state of SB.

The current state of SB is characterized by a rapid pace of research and development. In the past few years, there have been significant advances in the design and construction of synthetic biological systems. One of the most notable examples is the creation of the first synthetic cell, known as *Mycoplasma mycoides* JCVI-03. This cell was created by a team of researchers at the J. Craig Venter Institute (JCVI) and was the first time that a synthetic cell had been created. This achievement was a major milestone in the field of SB and has opened up new possibilities for the development of synthetic biological systems. Other notable achievements in SB include the creation of synthetic genes, the development of synthetic metabolic pathways, and the creation of synthetic cells that can perform specific tasks. These achievements have demonstrated the potential of SB to create biological systems that are more efficient, more reliable, and more predictable than natural biological systems.

3. Applications of SB.

SB has a wide range of applications, including medicine, agriculture, and environmental science. In medicine, SB is being used to develop new drugs, to create synthetic cells that can perform specific tasks, and to develop new diagnostic tools. In agriculture, SB is being used to create synthetic plants that can produce specific products, to develop new crops, and to create synthetic animals that can perform specific tasks. In environmental science, SB is being used to create synthetic organisms that can perform specific tasks, to develop new environmental sensors, and to create synthetic ecosystems. These applications demonstrate the potential of SB to revolutionize many areas of science and technology.

4. Challenges and opportunities.

There are several challenges and opportunities associated with SB. One of the major challenges is the need for a better understanding of the principles of biology. While there have been significant advances in our understanding of biology, there is still much that we do not know. This lack of understanding can make it difficult to design and construct synthetic biological systems. Another challenge is the need for better tools and techniques for the design and construction of synthetic biological systems. While there have been significant advances in these areas, there is still much that needs to be done. Finally, there are ethical and safety concerns associated with SB. These concerns include the potential for the creation of synthetic organisms that could be harmful to the environment or to human health, and the potential for the misuse of SB technology. Despite these challenges, there are also many opportunities associated with SB. These opportunities include the potential for the development of new drugs, the creation of new crops, and the development of new environmental sensors. SB has the potential to revolutionize many areas of science and technology, and it is an exciting field that is worth pursuing.

5. Conclusion.

SB is a rapidly growing field that has the potential to revolutionize many areas of science and technology. It is a multidisciplinary field that combines principles from biology, engineering, computer science, and chemistry to create artificial biological systems. These systems are designed to perform specific functions, such as producing a particular molecule, detecting a specific environmental condition, or performing a specific task. The goal of SB is to create biological systems that are more efficient, more reliable, and more predictable than natural biological systems. In this paper, we have discussed the current state of SB, its applications, and its future prospects. We have also discussed the challenges and opportunities associated with this field. SB is an exciting field that is worth pursuing, and it has the potential to revolutionize many areas of science and technology.

© 2011 The Authors. Journal of Internal Medicine © 2011 Blackwell Publishing Ltd

Publications related to ARETE project

Pub. 6





Collaborative Augmented Reality Tools for Behavioral Lessons

Ana Domínguez^{1(a)}, Álvaro Cabrero¹, Bruno Simões¹, Giuseppe Chiazze², Mariella Parolla², Marco Arrigo², Luciano Seta², Antonella Chifari², Crispino Tosto², Sui Lin Goei³, Eleni Mangina⁴, and Stefano Masneri⁵

¹ Vicomtech Foundation, Basque Research and Technology Alliance (BRTA), San Sebastián, Spain
adominguez@vicomtech.org

² National Research Council of Italy - Institute for Educational Technology, Palermo, Italy

³ Vrije Universiteit (VU), Amsterdam, Netherlands

⁴ University College of Dublin, Dublin, Ireland

⁵ University of the Basque Country (UPV/EHU), San Sebastián, Spain

Abstract. Multi-user Augmented Reality experiences have proven useful in increasing user engagement and facilitating the learning of new concepts. Thanks to the new generation of Augmented Reality frameworks for smartphones and tablets, their daily use in the classroom is becoming more common. In this paper, we present the work done towards the implementation of classroom-based behavioural lessons based on collaborative Augmented Reality environments. We have developed a library that enables multi-user Augmented Reality applications and have used it to implement some scenarios developed in the context Positive Behaviour Intervention and Support framework.

Keywords: Augmented Reality · Collaborative learning · Multi-user applications · Education · Behavioural lessons · PBIS

1 Introduction

Augmented Reality (AR) experiences are thriving and becoming increasingly important in the educational landscape to increase student motivation, engagement and knowledge retention. In fact, AR can be a breakthrough in education by transforming the entire learning experience. AR makes it possible to visualise computer-generated objects on a device while viewing the real world with a camera in real time. This allows students to learn in an immersive environment where 3D annotations and rich visual animations complement textbooks with interactive content. AR makes learning more engaging and effortless by simplifying concepts. All this makes students' attitudes more positive and increases their willingness to collaborate with other students and the teacher.

Given the above benefits and its increasing popularity, AR has become an active research topic. However, the possibilities for interaction and collaboration

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2023
A. E. Auer et al. (eds.) ICL 2022, LNCS 1550, pp. 102–109, 2023.
doi.org/10.1007/978-3-031-26876-2_10

Domínguez, A., Cabrero, Á., Simões, B., Chiazze, G., Farella, M., Arrigo, M., Seta, L., Chifari, A., Tosto, C., Goei, S.L., Mangina, E. & Masneri, S. (2023). Collaborative Augmented Reality Tools for Behavioral Lessons. In 25th International Conference on Interactive Collaborative Learning, ICL 2022 (pp. 102-109). Springer Science and Business Media Deutschland GmbH. https://doi.org/10.1007/978-3-031-26876-2_10

Collaborative Augmented Reality

Ana Domínguez¹, Álvaro Cabrero¹, Bruno Simões¹, Giuseppe Chiazze², Mariella Parolla², Marco Arrigo², Luciano Seta², Antonella Chifari², Crispino Tosto², Sui Lin Goei³, Eleni Mangina⁴, and Stefano Masneri⁵

¹ Vicomtech Foundation, Basque Research and Technology Alliance (BRTA), San Sebastián, Spain
adominguez@vicomtech.org

² National Research Council of Italy - Institute for Educational Technology, Palermo, Italy

³ Vrije Universiteit (VU), Amsterdam, Netherlands

⁴ University College of Dublin, Dublin, Ireland

⁵ University of the Basque Country (UPV/EHU), San Sebastián, Spain

Abstract. Multi-user Augmented Reality experiences have proven useful in increasing user engagement and facilitating the learning of new concepts. Thanks to the new generation of Augmented Reality frameworks for smartphones and tablets, their daily use in the classroom is becoming more common. In this paper, we present the work done towards the implementation of classroom-based behavioural lessons based on collaborative Augmented Reality environments. We have developed a library that enables multi-user Augmented Reality applications and have used it to implement some scenarios developed in the context Positive Behaviour Intervention and Support framework.

Keywords: Augmented Reality · Collaborative learning · Multi-user applications · Education · Behavioural lessons · PBIS

1 Introduction

Augmented Reality (AR) experiences are thriving and becoming increasingly important in the educational landscape to increase student motivation, engagement and knowledge retention. In fact, AR can be a breakthrough in education by transforming the entire learning experience. AR makes it possible to visualise computer-generated objects on a device while viewing the real world with a camera in real time. This allows students to learn in an immersive environment where 3D annotations and rich visual animations complement textbooks with interactive content. AR makes learning more engaging and effortless by simplifying concepts. All this makes students' attitudes more positive and increases their willingness to collaborate with other students and the teacher.

Given the above benefits and its increasing popularity, AR has become an active research topic. However, the possibilities for interaction and collaboration

Publications related to ARETE project

Pub. 6

Collaborative Antismoking Media Tools

[illegible]

where $\mathbf{y}$ is the vector of observed values, $\mathbf{X}$ is the matrix of explanatory variables, β is the vector of parameters to be estimated, and ϵ is the vector of error terms. The error term represents the unexplained variation in the dependent variable. The model is estimated using the method of ordinary least squares (OLS), which minimizes the sum of the squared residuals. The resulting parameter estimates are used to predict the values of the dependent variable for a given set of explanatory variables.

the authors find interesting differences between the two types of capital. In the case of fixed capital, the authors find that the more capital-intensive the economy, the more capital-intensive the technology. In the case of human capital, the authors find that the more capital-intensive the economy, the less capital-intensive the technology. This is a very interesting finding, and it suggests that the two types of capital may have different effects on the economy. The authors also find that the more capital-intensive the economy, the more capital-intensive the technology. This is a very interesting finding, and it suggests that the two types of capital may have different effects on the economy.

Publications related to ARETE project

Pub. 6

scientific data

OPEN

DATA DESCRIPTOR

Dataset of user interactions across four large pilots on the use of augmented reality in learning experiences

Ana Domínguez^{1,2*}, Guillermo Pachó³, Lisa Bowers⁴, Fridolin Wild⁵, Sarah Alcock⁶, Giuseppe Chiazese⁷, Mariella Farella⁸, Marco Arrigo⁹, David Ross¹, Rita Treacy¹, Darya Yegorina¹, Eleni Mangina¹⁰ & Stefano Masneri¹¹

Augmented Reality in education can support students in a wide range of cognitive tasks—fostering understanding, remembering, applying, analysing, evaluating, and creating learning-relevant information more easily. It can help keep up engagement, and it can render learning more fun. Within the framework of a multi-year investigation encompassing primary and secondary schools across Europe, the ARETE project developed several Augmented Reality applications, providing tools for user interaction and data collection in the education sector. The project developed innovative AR learning technology and methodology, validating these in four comprehensive pilot studies, in total involving more than 2,900 teachers and students. Each pilot made use of a different Augmented Reality application covering specific subjects (English literacy skills, Mathematics and Geography, Positive Behaviour, plus, additionally, an Augmented Reality authoring tool applied in a wide range of subjects). In this paper, we introduce the datasets collected during the pilots, describe how the data enabled the validation of the technology, and how the approach chosen could enhance existing augmented reality applications in data exploration and modelling.

Background & Summary

By applying Augmented Reality (AR) technology in an education setting, new applications emerge which can positively affect the process of teaching and learning for students in real-world situations^{1–4}. Like all technology-enhanced learning, AR in education creates data trails⁵ from which teachers and researchers can glean insight into actual learner behaviour. For this, linking AR interactions with all Learning Analytics is paramount⁶ for tracking and analysis, drawing from a variety of tools and approaches.

Most notably, up until recently this typically would involve SCORM⁷, a set of technical standards for e-learning products, which enable tracking of learners' experiences at a rather high level. While the SCORM provided granularity was sufficient for decades, today an enormous rise for educational data mining, data scientists and pedagogues are using machine learning techniques to analyse the learning experience, requiring more fine-grained, more in-depth, and larger volumes of data.

The Experience Application Programming Interface (sAPI, <https://github.com/andrea/sapi-spec>) is an open e-learning specification, which allows recording nearly every type of learning activity. For example, it can track which buttons the user clicks or how long the learner spent reading a page. In order to do that, it collects data on the learner activity in real-time, storing it in an easy-to-read semantic string composed of three main constituents: actor, verb and object (for example, 'Sally played the instructional video').

Previously, the sAPI has already been used as a data collection system in AR, such as in 'AugmentiaTM'⁸ which used learning activity data to display it as an interactive visualisation. It was also used in the area of

¹Vicomtech Foundation, Basque Research and Technology Alliance (BRTA), Digital Media, San Sebastián, 20099, Spain. ²The Open University, Milton Keynes, MK7 6AA, United Kingdom. ³Consiglio Nazionale delle Ricerche, Pinerolo, 10146, Italy. ⁴Ward-North Learning, Newry, BT4 5DZ, Ireland. ⁵Cineca, Dublin, D15 DPT1, Ireland. ⁶UCD, Dublin, Dublin 4, Ireland. ⁷Computer Languages and Systems Department, University of the Basque Country UPV/EHU, San Sebastián, 20011, Spain. ⁸✉ e-mail: adominguez@vicomtech.org

Domínguez, A., Pachó, G., Bowers, L., Wild, F., Alcock, S., Chiazese, G., Farella, M., Arrigo, M., Ross, D., Treacy, T., Yegorina, D., Mangina, E. & Masneri, S. (2023). Dataset of user interactions across four large pilots on the use of augmented reality in learning experiences. *Sci Data* 10, 823 (2023). <https://doi.org/10.1038/s41597-023-02743-6>

Publications related to ARETE project

Pub. 6

Work-in-progress—ARITE: An Interactive Educational System using Augmented Reality

Referee Name(s) Referee 1 Referee 2 Referee 3	Referee Name(s) Referee 4 Referee 5 Referee 6	Referee Name(s) Referee 7 Referee 8 Referee 9
Review The authors have presented a paper that describes the development of ARITE, an interactive educational system using augmented reality. The paper is well written and easy to read. The authors have provided a clear description of the system and its components. The paper is well organized and the authors have provided a clear description of the system and its components. The paper is well organized and the authors have provided a clear description of the system and its components.	Review The authors have presented a paper that describes the development of ARITE, an interactive educational system using augmented reality. The paper is well written and easy to read. The authors have provided a clear description of the system and its components. The paper is well organized and the authors have provided a clear description of the system and its components. The paper is well organized and the authors have provided a clear description of the system and its components.	Review The authors have presented a paper that describes the development of ARITE, an interactive educational system using augmented reality. The paper is well written and easy to read. The authors have provided a clear description of the system and its components. The paper is well organized and the authors have provided a clear description of the system and its components. The paper is well organized and the authors have provided a clear description of the system and its components.
Authors' Note The authors would like to thank the following people for their help and support during the development of ARITE: [List of names]	Authors' Note The authors would like to thank the following people for their help and support during the development of ARITE: [List of names]	Authors' Note The authors would like to thank the following people for their help and support during the development of ARITE: [List of names]

ARITE (Augmented Reality Interactive Teaching Environment) is a software system that provides a virtual environment for learning. It is designed to be used in a classroom setting, where students can interact with the system and learn about various topics. The system is designed to be user-friendly and easy to use, and it provides a variety of features that make learning more engaging and effective.

Key features of ARITE include:

- Interactive Learning:** Students can interact with the system and learn about various topics. The system is designed to be user-friendly and easy to use, and it provides a variety of features that make learning more engaging and effective.
- Virtual Environment:** The system provides a virtual environment where students can learn about various topics. This environment is designed to be immersive and engaging, and it provides a variety of features that make learning more effective.
- Customizable Content:** The system allows users to customize the content to meet their needs. This includes the ability to create custom lessons, quizzes, and other learning materials.
- Assessment Tools:** The system includes a variety of assessment tools that allow users to track student progress and performance. These tools include quizzes, tests, and other assessment methods.
- Reporting Tools:** The system includes a variety of reporting tools that allow users to generate reports on student performance. These reports can be used to identify areas where students are struggling and to provide targeted support.

Conclusion: ARITE is a powerful tool for learning that provides a variety of features that make learning more engaging and effective. It is designed to be user-friendly and easy to use, and it provides a variety of features that make learning more effective. The system is a valuable tool for educators and students alike, and it is a great way to enhance the learning experience.

References:

1. [Reference 1]
2. [Reference 2]
3. [Reference 3]
4. [Reference 4]
5. [Reference 5]
6. [Reference 6]
7. [Reference 7]
8. [Reference 8]
9. [Reference 9]
10. [Reference 10]

Keywords: ARITE, Augmented Reality, Interactive Teaching Environment, Virtual Environment, Customizable Content, Assessment Tools, Reporting Tools.

Index Terms: ARITE, Augmented Reality, Interactive Teaching Environment, Virtual Environment, Customizable Content, Assessment Tools, Reporting Tools.

Authors' Address: [Address]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website: [Website]

Authors' LinkedIn: [LinkedIn]

Authors' Twitter: [Twitter]

Authors' Facebook: [Facebook]

Authors' YouTube: [YouTube]

Authors' Instagram: [Instagram]

Authors' Snapchat: [Snapchat]

Authors' TikTok: [TikTok]

Authors' Twitch: [Twitch]

Authors' Discord: [Discord]

Authors' Slack: [Slack]

Authors' WhatsApp: [WhatsApp]

Authors' Telegram: [Telegram]

Authors' Messenger: [Messenger]

Authors' Email: [Email]

Authors' Phone: [Phone]

Authors' Fax: [Fax]

Authors' Website:</

Collective Amnesia: Redefining the Holocaust

The image shows the front cover of the journal 'scientific data'. At the top left is the 'scientific data' logo. Below it, the word 'OPEN' is in a blue box. The main title is 'Dataset of user interactions across four large pilots on the use of augmented reality in learning experiences'. Below the title, the authors are listed: 'Rachael Macdonald¹, Stephen Houghton¹, Catherine Houghton¹, Andrew Houghton¹, David Houghton¹, David Houghton¹'. The journal name 'SCIENTIFIC DATA' is at the bottom. The cover features a dark background with a grid of small, colorful squares.

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE ART

Overview
Methodology
Results
Discussion

RESEARCH RESULTS

Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

01

SLR data analysis: https://github.com/Stocastico/AR_SLR_Paper

02

Orkestra {
Client: <https://github.com/tv-vicomtech/orkestraClient>
Server: <https://github.com/tv-vicomtech/orkestraServer>
Unity: <https://github.com/tv-vicomtech/orkestraLibUnity>

03

cleAR {
Definition & PoC: https://github.com/Stocastico/ARchitecture_paper
xAPI: https://github.com/Stocastico/xapi_analysis

04

Validation {
ARoundTheWorld: <https://github.com/tv-vicomtech/ARoundtheworld>
Data analysis: https://github.com/Stocastico/Evaluation_paper

INTRODUCTION

Motivation

Objectives

Methodology

Contributions

STATE OF THE
ART

Overview

Methodology

Results

Discussion

RESEARCH
RESULTSInteroperable
architecture for
collaborative
AR in educationValidation of the
proposed
architecture

CONCLUSIONS

Conclusions

Publications

Future work

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

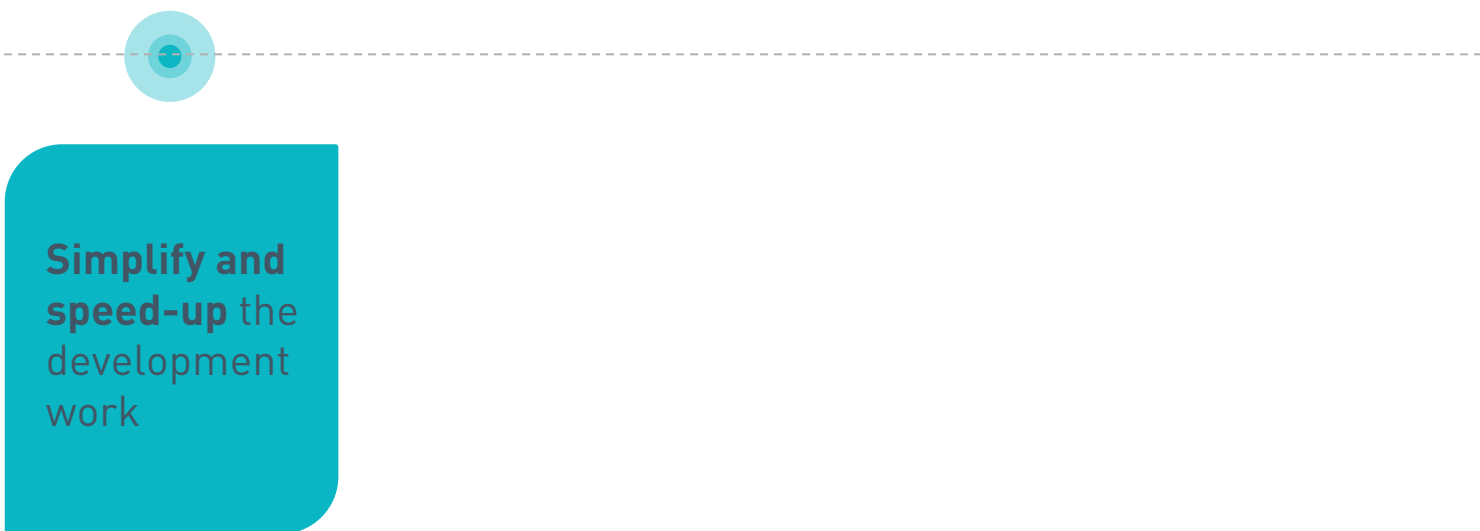
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications

Future work



**Simplify and
speed-up** the
development
work

INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

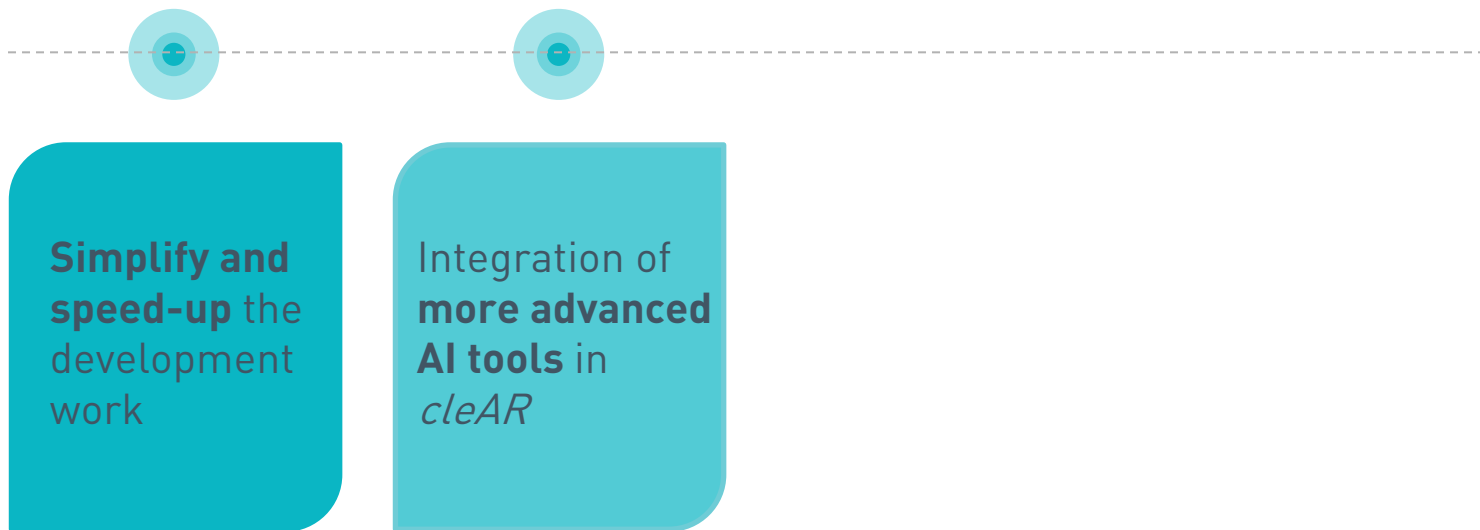
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications

Future work



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

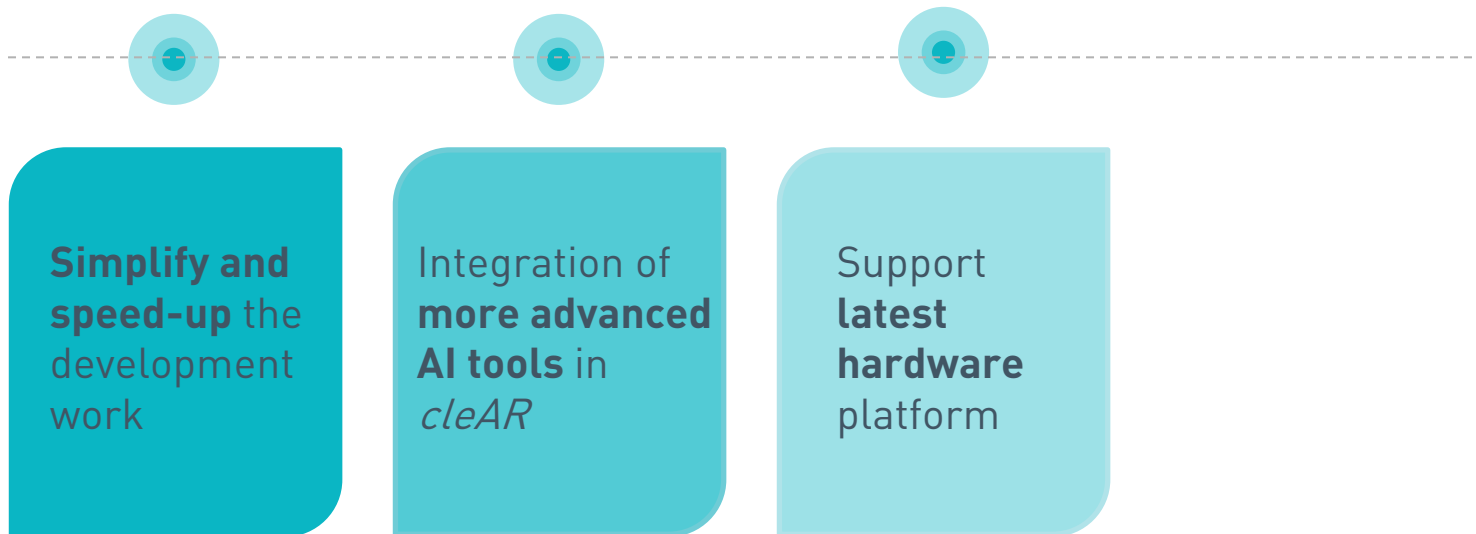
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications

Future work



INTRODUCTION

Motivation
Objectives
Methodology
Contributions

STATE OF THE
ART

Overview
Methodology
Results
Discussion

RESEARCH
RESULTS

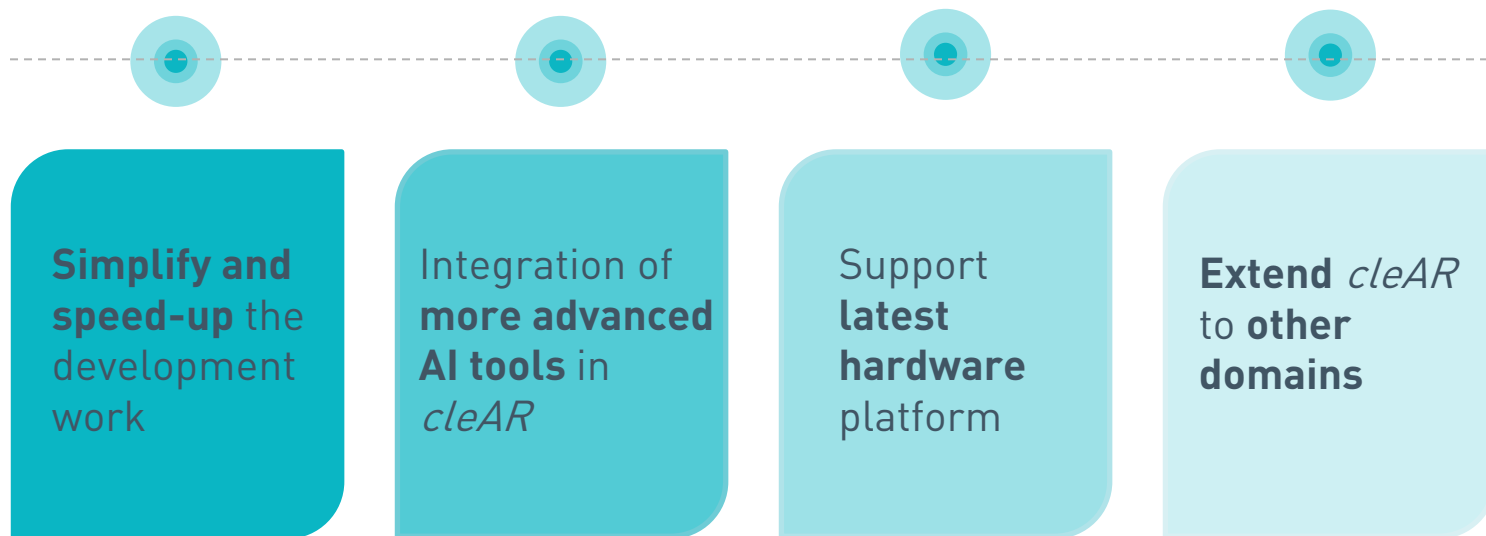
Interoperable
architecture for
collaborative
AR in education

Validation of the
proposed
architecture

CONCLUSIONS

Conclusions
Publications

Future work



A Novel Architecture for Collaborative Augmented Reality Experiences for Education

By Stefano Masneri

Supervised by Dr. Ana Arruarte Lasa and Dr. Mikel Zorrilla Berasategi

Donostia – San Sebastián

Friday, 8 March 2024